Controlled Inversion of Image-Based Time-Series Encodings: Operational Decoders for GAF, RP, and MTF Representations

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Image-based encodings of time series — Recurrence Plots (RP), Markov Transition Fields (MTF), and the summation (GASF) and difference (GADF) variants of Gramian Angular Fields — expose temporal structure to image-domain models [9, 10]. This work does not evaluate generative augmentation; it studies the prerequisite inverse problem: whether images from known encoders can be mapped back to triaxial accelerometer windows with controlled loss. We contribute three operational decoders with, to our knowledge, no prior fully specified counterpart — a signed-RP decoder that recovers the two-dimensional adjacent-state geometry — correcting the inverse normalisation and resolving the multidimensional-scaling orthogonal ambiguity through a consecutive-state overlap constraint and the Lu–Tong convention — where Hirata, Horai and Aihara [7] stopped at the ambiguous multidimensional-scaling embedding; a closed-form, multi-pivot, boundary-anchored GADF decoder with a fixed arcsin branch, image-derived anchor and sign convention; and a Fiedler-vector spectral MTF decoder — alongside a GASF diagonal inverse [11] retained as a controlled prior-art baseline.

Under a unified four-lens protocol (visual, pointwise error, polynomial trend, and six-classifier original-trained robustness) on WISDM windows ($N = 128$), evaluated under both window-level and grouped subject-out 3-fold splits, GASF and GADF are both near-original: GADF shows no detectable classification difference from GASF or the raw signal (paired McNemar $p_{\text{Holm}} = 1.0$ across all six classifiers) and yields the lower observed pointwise error (global MAE 0.0156 vs 0.0202). We therefore recommend $\{\text{GASF}, \text{GADF}\}$ as the reconstruction-oriented GAF baseline, without claiming formal information preservation. RP is likewise near-lossless once its inverse normalisation is corrected (global MAE 0.0148, Pearson ≈ 1.0), and its reconstructions match the original-signal classification accuracy ($\Delta\text{acc} \approx 0$ across the six classifiers), leaving MTF as the only lossy decoder — valuable as a compact classifier-oriented representation rather than a faithful inverse. Because every decoder assumes known encoder settings, this is controlled reconstruction — a necessary step toward, not a proof of, generative augmentation.

Keywords: Markov Transition Fields, Recurrence Plots, Gramian Angular Fields, Gramian Angular Difference Field, Time Series Reconstruction, Image Encoding, Accelerometer Data, Machine Learning.

1. Introduction

Time series data play a crucial role in fields ranging from finance and healthcare to human activity recognition. Despite their importance, many applications are constrained by scarce labelled data, which has motivated interest in data augmentation and, more broadly, in alternative representations that can leverage advanced methods from adjacent domains — most notably the rich domain of image processing.

A common approach is to convert time-series signals into image-based representations. Techniques such as Markov Transition Fields (MTF), Recurrence Plots (RP), and Gramian Angular Fields (GAF) transform a one-dimensional sequence into a two-dimensional matrix whose pixel (i, j) encodes a relation between two time indices. For triaxial accelerometer windows, the three axes are encoded independently and then treated as image channels. This construction

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allows image analysis and generation models to operate on temporal structure that is often difficult to expose directly in the raw signal.

Parallel to these developments, diffusion models have gained prominence in image generation. Their ability to synthesize highly structured images motivates the use of image representations for time-series augmentation, but this paper does not train or evaluate a diffusion or GAN model. We focus on the necessary inverse problem that must be solved first: if an image is produced by a known time-series encoder, can it be reconstructed back into the signal domain with controlled loss?

Our approach addresses this inverse problem by proposing dedicated reconstruction strategies for RP, MTF, and GAF-based encodings, and by evaluating the completed decoders under a unified framework that includes visual inspection, reconstruction metrics, polynomial trend comparison, and downstream classification. The overall workflow is summarized in Fig. 1. Experiments on the WISDM Smartphone and Smartwatch Activity and Biometrics Dataset [14] show that the evaluated encodings preserve different aspects of the original signals and therefore support different use cases.

A central narrative choice in this paper is to separate *controlled reconstruction* from *unconstrained generative augmentation*. The experiments decode images produced by known encoders with known encoder settings and, when physical units are reported, use per-window min–max calibration recorded during encoding. This setting is the necessary first test of whether an image representation contains enough information to return to the time domain. It is not yet a claim that arbitrary images produced by a generative model will be physically plausible or reconstructable without additional calibration.

The paper is organized around three questions: what information each image encoding preserves, how that information can be mapped back to the time domain, and whether the reconstructed signals remain useful for downstream activity recognition. The main contributions are:

- a common operator-level formulation $M_{ij} = \omega(\xi(x_i), \xi(x_j))$ that, in the discussion, lets us read signed RP and GADF as structurally adjacent parametrisations of directional pairwise differences — GADF lying in the antisymmetric zero-diagonal matrix space by construction, signed RP approximating it without exact antisymmetry under the Lu–Tong sign rule;
- **a novel RP decoder** that recovers the full two-dimensional adjacent-state geometry from the signed Lu–Tong RP left only partially inverted by Hirata–Horai–Aihara [7]: it corrects the inverse normalisation (decoding the image with the RP matrix's own range), recovers the states by classical multidimensional scaling, and resolves the residual orthogonal transformation with a consecutive-state overlap constraint and the signed-RP convention — the ablation of Section 6.6 shows the inverse-normalisation correction and the signed orientation selection are the two decisive components, together making RP reconstruction near-lossless (MAE ≈ 0.015, Pearson ≈ 1.0);
- **a novel restricted-branch, multi-pivot, boundary-anchored GADF decoder** that operationalises the earlier observation by Wang and Oates [9] that GADF can admit an accurate inverse map under restricted scaling: the contribution is a fully specified closed-form decoder with an explicit arcsin branch condition, image-derived angular anchor, fixed sign convention, optional min–max calibration for physical-scale recovery, and empirical evaluation under the same protocol as RP, MTF, and GASF, so it should not be read as a generic inverse for unconstrained GADF images;
- **a novel Fiedler-vector MTF decoder**, for which we are not aware of a prior explicit MTF-to-time-series decoder: rows of the field are clustered into Q bins, the bin transition matrix is recovered, and the bin ordering is fixed by the Fiedler vector of the corresponding graph Laplacian; because MTF is quantised and many-to-one, the output is a representative compatible quantile trajectory rather than a bijective inverse;
- a GASF inversion pipeline that extends the diagonal-inverse line of work [11] with explicit handling of normalisation and quantisation, kept in the comparison as a controlled prior-art baseline rather than as a novel decoder;
- an explicit account of the assumptions that make each inverse map exact, approximate, or heuristic, and the encoder settings or optional calibration that each one requires;
- an evaluation design that separates pointwise fidelity, trend-level similarity, and classification utility, so that good classification is not mistaken for accurate signal reconstruction;
- a fully comparable evaluation of all four encodings (RP, MTF, GASF, GADF) under the same protocol – same fold, same six-classifier panel, same pointwise and polynomial-trend metrics, both stratified window-level and grouped subject-out 3-fold splits.

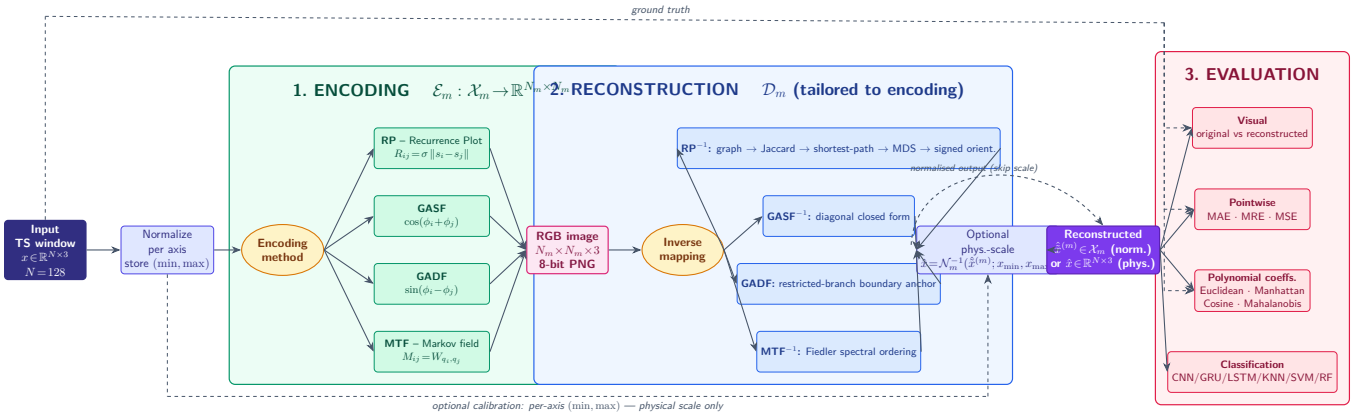


Figure 1: Controlled image-based time-series reconstruction workflow. Solid arrows show the signal/image path from the triaxial window $x \in \mathbb{R}^{N\times 3}$ through per-axis encoding ($\mathcal{E}_m : \mathcal{X}_m \to \mathbb{R}^{N_m \times N_m}$, with $N_m = N$ for GASF/GADF/MTF and $N_m = N-1$ for RP) and method-tailored inversion ($\mathcal{D}_m$), which returns a normalised reconstruction $\hat{\tilde{x}}^{(m)} \in \mathcal{X}_m$. The final block is dual: the decoder output can be consumed directly in the normalised domain, or passed through the *optional* physical-scale recovery $\mathcal{N}_m^{-1}$ to obtain $\hat{x}$ in accelerometer units. Dashed arrows show auxiliary information recorded during encoding: the per-axis $(\min, \max)$ calibration, used *only* when physical-scale recovery is requested, and the original window used as ground truth for visual, pointwise, and trend metrics.

2. Related Work and Encodings

This section fixes the notation and summarizes the encoding mechanisms needed for the inverse problem. We distinguish between the image construction rule, which is part of the prior literature, and the reconstruction rule, which is where this paper contributes or identifies an explicit gap. For clarity, GAF denotes the family, whereas GASF and GADF denote the summation and difference variants, respectively.

All four encodings can be written as per-axis operators $\mathcal{E}_m : \mathcal{X}_m \to \mathbb{R}^{N_m \times N_m}$, but they preserve different kinds of information. RP encodes geometric recurrences in a phase-space embedding, GASF and GADF encode angular relations after polar normalization, and MTF encodes transition probabilities between quantised states. This distinction is important for the rest of the paper: a representation that is useful for classification is not necessarily pointwise invertible, and a representation that is highly invertible under controlled conditions may still require metadata that are difficult to obtain for generated images.

Table 1: Operator-level view of the four image encodings.

Method	**Operator**	**Main Information Preserved**	**Inverse Character**
RP	$\text{sign}(i,j)\,\|s_i - s_j\|$	Phase-space geometry and recurrence	Approximate, graph/MDS based
GASF	$\cos(\phi_i + \phi_j)$	Absolute angular magnitude via the diagonal	Closed form under a fixed branch
GADF	$\sin(\phi_i - \phi_j)$	Relative angular differences	Closed form with restricted scaling and anchor
MTF	W_{q_i,q_j}	Markov transition statistics between bins	Heuristic, quantised reconstruction

Table 2 positions each decoder against the closest prior reconstruction work and states the specific gap it closes, so that the novelty claims made throughout the paper can be checked against a single reference point. Three of the four decoders (RP, GADF, MTF) have, to the best of our knowledge, no prior fully specified time-series inverse; GASF is the exception and is kept only as a controlled prior-art baseline.

Table 2: Closest prior reconstruction work per encoding and the gap this paper closes. "Prior inverse to the time series?" asks whether earlier work recovered the signal itself, not merely restored or classified the image.

Method	Closest prior reconstruction work	Prior inverse to the time series?	What this paper adds
RP	Hirata–Horai–Aihara [7]; Thiel [5]; Tanio et al. [6]	Partial: recovers a state ordering up to a reflection, stopping at MDS	Corrected inverse normalisation (RP-range decode) with two-dimensional MDS state recovery, consecutive-state overlap alignment, and signed-RP orientation selection — near-lossless (MAE ≈ 0.015, Pearson ≈ 1.0)
GASF	Huamin et al. [11] (diagonal half-angle)	Yes, from the main diagonal	Re-implemented as a controlled prior-art baseline with explicit normalisation/quantisation handling (not claimed as novel)
GADF	Wang and Oates [9] note invertibility under $[0, 1]$ scaling	Stated but not specified: their practical route goes through the GASF diagonal	Explicit closed-form multi-pivot, boundary-anchored decoder with fixed arcsin branch, image-derived anchor and sign convention (min–max only for optional physical-scale recovery)
MTF	No prior MTF→series decoder; MRF work [12, 13] restores images, not series	No	Fiedler-vector spectral bin-ordering yielding a representative compatible quantile trajectory

2.1. Recurrence Plot

2.1.1. Image Construction

Recurrence plots were introduced by Eckmann et al. [1] and are commonly interpreted through phase-space reconstruction ideas developed for nonlinear dynamical systems [2, 3]. A scalar time series can be embedded as delay vectors

$$\mathbf{s}_t = (x_t, x_{t+\tau}, \ldots, x_{t+(m-1)\tau}) \in \mathbb{R}^m,$$

where m is the embedding dimension and τ is the delay. In this work, we use the first-order embedding $m = 2$, $\tau = 1$, so $s_i = (x_i, x_{i+1})$. The classical recurrence matrix is then defined as

$$\mathbf{R}_{i,j}(\epsilon) = \boldsymbol{\theta}(\epsilon - \|s_i - s_j\|), \quad i, j = 1, \ldots, N-1, \tag{1}$$

where ϵ is a threshold that determines the neighborhood size, $\boldsymbol{\theta}$ is the Heaviside step function, and $\|\cdot\|$ denotes a norm. Choosing an appropriate norm is crucial for constructing a meaningful recurrence plot; the most commonly used norms are L_1, L_2, and L_∞ [4].

One limitation of this definition is that the distance from x_i to x_j equals that from x_j to x_i, yielding a symmetric matrix with respect to the main diagonal. Consequently, the mapping is not bijective: a time series and its inverted version produce identical images, meaning that decoding the image would yield two possible time-series representations. To overcome this issue, we adopt the alternative definition proposed by Jianjie Lu and Kai-Yu Tong [4], which addresses the bijective image representation problem and is formulated as follows:

Consider a time series $x \in \mathbb{R}^{1\times N}$, where x_j denotes the j-th element of x. The states are then defined as

$$s_j = \big(x_j,\, x_{j+1}\big), \quad \text{with } s_j \in \mathbb{R}^2,$$

so that the recurrence plot (RP) is constructed as a matrix $R \in \mathbb{R}^{(N-1)\times(N-1)}$.

A basic formulation of the RP is given by

$$R_{m,n} = \|s_m - s_n\|, \quad m, n = 1, \ldots, N-1, \tag{2}$$

where $\|\cdot\|$ denotes a chosen norm (e.g., L_1, L_2, or L_∞).

To incorporate directional information, we calculate a sign based on the differences between state vectors. Throughout this work we use the fixed reference direction $v = (1, 1)^\top/\sqrt{2}$. For off-diagonal entries, define

$$\gamma_{mn} = \frac{(s_m - s_n)\cdot v}{\|s_m - s_n\|\,\|v\|}, \qquad m \neq n.$$

The sign convention assigns a negative value only when the off-diagonal displacement vector forms an angle greater than $3\pi/4$ with respect to v:

$$\operatorname{sign}(m, n) = \begin{cases} 1, & \text{if } m = n, \\ -1, & \text{if } m \neq n \text{ and } \gamma_{mn} < \cos\left(\dfrac{3\pi}{4}\right), \\ 1, & \text{otherwise,} \end{cases} \tag{3}$$

where $\|v\| = 1$. The diagonal convention avoids the undefined ratio at $m = n$; since $R_{m,m} = 0$ after multiplication by the distance, the chosen diagonal sign has no numerical effect. This convention is used consistently during both RP encoding and the reconstruction-orientation check.

Finally, the recurrence plot is redefined as

$$R_{m,n} = \text{sign}(m, n) \cdot \|s_m - s_n\|, \tag{4}$$

which is the definition we use to encode the RP as an image after normalizing the values and mapping them to RGB.

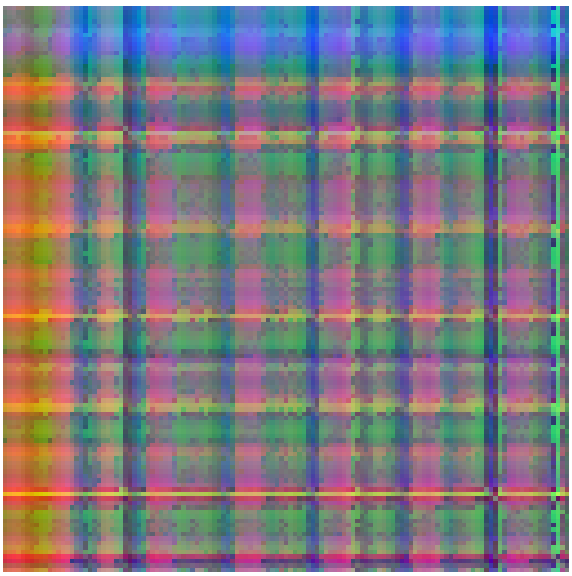


Figure 2: Example of a Recurrence Plot.

The adjacent-state construction produces $N - 1$ embedded states rather than N scalar samples. The information loss is not merely a count mismatch, since the overlapping states still contain all samples in principle; it arises because the RP stores pairwise distances or thresholded recurrences, which are invariant to transformations such as translation, rotation, reflection, and, depending on normalization, scale. The signed Lu–Tong convention used here adds an orientation cue, but it does not make the RP representation fully bijective.

2.1.2. Time Series Reconstruction

There is no direct inverse transformation for converting time series into recurrence plots according to the literature. However, several approaches have been proposed over time, primarily because this is one of the oldest methods for reconstruction. Notable contributions include those by M. Thiel [5] and Tanio et al. [6]. In particular, we highlight the decoding method by Hirata, Horai, and Aihara [7], which, as shown later, is adapted here to our signed RP definition (see Equation 4). This method applies to both continuous and binary recurrence plots.

1. **Starting from a Recurrence Plot:** Let $z(i)$ denote the state at time i (with $1 \leq i \leq n$). Define a function c that takes two states and indicates whether they are sufficiently close; it returns 1 if they are close and 0 otherwise, according to a predefined threshold. Thus, the recurrence plot can be represented as a binary cost matrix:
$$\{(i, j) : c(z_i, z_j) = 1, \quad i, j = 1, 2, \ldots, n\}.$$
2. **Graph Construction:** Construct a graph in which an edge is drawn between nodes i and j if $c(z_i, z_j) = 1$.
3. **Edge Weight Assignment:** For each state z_i, define the set
$$G_i = \{j : c(z_i, z_j) = 1\},$$
which represents the indices of time where state z_i is close to other states. Then, assign a weight to each edge between i and j as follows:
$$w(i, j) = 1 - \frac{|G_i \cap G_j|}{|G_i \cup G_j|}.$$
4. **Shortest Path Calculation:** Compute the shortest path distances in the graph using Dijkstra's algorithm [8]. For larger graphs, an alternative method for finding the shortest path can be employed.

5. **Multidimensional Scaling:** Finally, apply a multidimensional scaling technique that orders the set of points using the component corresponding to the largest eigenvalue. We use *classical* (Torgerson) MDS, i.e. the closed-form double-centering eigendecomposition of $-\frac{1}{2}JD^2J$ with $J = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$, rather than the SMACOF iterative optimiser. For an exactly Euclidean distance matrix, classical MDS gives an exact embedding when all required positive-eigenvalue dimensions are retained; here we keep only the leading coordinate, a deterministic one-dimensional approximation that still avoids the stochastic initialisation dependence of an iterative SMACOF solver.

Although this graph-based method can recover a consistent state ordering, it has a notable drawback, also discussed in [7]: multidimensional scaling may return a mirror-symmetric reconstruction. In our WISDM experiments, this ambiguity was observed before applying the signed-RP orientation check described below; therefore, the observed ambiguity should be understood as an empirical diagnostic of this implementation rather than as a parameter from the cited method.

2.2. Gramian Angular Field

2.2.1. Image Construction

Introduced by Wang, Z. and Oates, T. in *Imaging Time-Series to Improve Classification and Imputation* [9], the Gramian Angular Field (GAF) technique represents time series values in polar coordinates. To do so, it is essential to rescale the time series values to an appropriate interval either to $[-1, 1]$ to map to angles in $[0, \pi]$, or to $[0, 1]$ to map to angles in $[0, \pi/2]$.

Once the time series is rescaled, the encoding is defined using the arccosine and the radius. Specifically, for the interval $[0, \pi]$ the definitions are:

$$\begin{cases} \phi = \arccos(\tilde{x}_i), & -1 \leq \tilde{x}_i \leq 1, \quad \tilde{x}_i \in \tilde{X}, \\ r = \frac{t_i}{N}, & t_i \in \mathbb{N}, \end{cases} \tag{5}$$

and for the interval $[0, \pi/2]$ they become:

$$\begin{cases} \phi = \arccos(\tilde{x}_i), & 0 \leq \tilde{x}_i \leq 1, \quad \tilde{x}_i \in \tilde{X}, \\ r = \frac{t_i}{N}, & t_i \in \mathbb{N}. \end{cases} \tag{6}$$

According to Wang and Oates [9], the polar transformation preserves temporal ordering through the radius and maps normalized amplitudes to angles through a monotonic arccosine branch. This does not mean that every GAF matrix has the same inverse properties. GASF and GADF use different pairwise angular operators, so the reconstruction argument must be derived separately for each case: GASF exposes amplitude information on the diagonal, whereas GADF stores only relative angular differences and has a zero diagonal.

Finally, two approaches to obtain a GAF are defined:

$$\begin{aligned} G^{\mathrm{S}}_{ij} &= \cos(\phi_i + \phi_j) \\ &= \tilde{x}_i\tilde{x}_j - \sqrt{1-\tilde{x}_i^2}\sqrt{1-\tilde{x}_j^2}, \end{aligned} \tag{7}$$

$$\begin{aligned} G^{\mathrm{D}}_{ij} &= \sin(\phi_i - \phi_j) \\ &= \sqrt{1-\tilde{x}_i^2}\tilde{x}_j - \tilde{x}_i\sqrt{1-\tilde{x}_j^2}. \end{aligned} \tag{8}$$

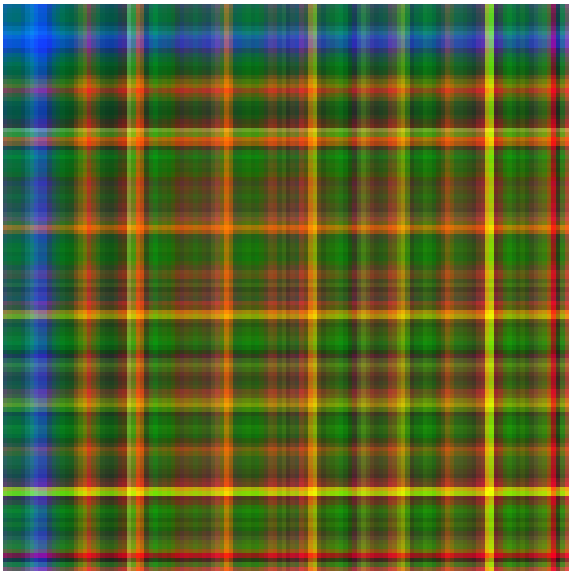


Figure 3: Example of a Gramian Angular Field.

2.2.2. Time Series Reconstruction

The two GAF variants require different reconstruction arguments. GASF has a known diagonal-based inverse under the appropriate normalization range, whereas GADF cannot be treated by simply reusing the same diagonal argument. We therefore separate the current GASF baseline from the GADF off-diagonal reconstruction developed below.

GASF reconstruction. In the original GAF paper [9], the possibility of recovering the original time series is connected to the structure of the Gramian representation. In subsequent literature, including a later paper by the same authors [10], the practical reconstruction route is discussed through the Gramian Angular Summation Field (GASF) main diagonal.

For GASF inversion, we use the main-diagonal argument. The diagonal half-angle reconstruction is already noted in the original GAF paper by Wang and Oates [9]; we follow the explicit normalization conditions later stated by Tao, Deng, and Xiao [11], under which the diagonal can be used for reconstruction. Under those conditions, the normalized time series can be reconstructed by observing the main diagonal of a GASF. Specifically, since the diagonal elements satisfy

$$G^{\mathrm{S}}_{ii} = \cos(2\phi_i),$$

one can recover the normalized sample using the half-angle identity:

$$\tilde{x}_i = \cos\phi_i = \sqrt{\frac{G^{\mathrm{S}}_{ii} + 1}{2}}, \quad \phi_i \in \left[0, \frac{\pi}{2}\right].$$

It is important to note that the main-diagonal inversion is not unique for the interval $[-1, 1]$, which maps to angles in $[0, \pi]$. In that case,

$$G^{\mathrm{S}}_{ii} = \cos(2\phi_i) = 2\tilde{x}_i^2 - 1,$$

so the half-angle identity recovers only $|\tilde{x}_i|$, not the sign of $\tilde{x}_i$. This limitation applies to the diagonal-only inversion rule used here; it does not constitute a proof that no reconstruction could be obtained from the full GASF matrix or from additional side information.

Wang and Oates state that the GADF of $[0, 1]$-rescaled data has an accurate inverse map [9]. However, their reconstruction discussion develops the practical inverse through the GASF diagonal and does not provide an explicit GADF decoding algorithm. In particular, a GADF decoder must resolve the branch of arcsin, fix the otherwise free global angular offset implied by the difference-only form $\sin(\phi_i - \phi_j)$, keep the sign convention fixed, and map the recovered angular trajectory back to normalised amplitudes; recovery of physical accelerometer units is an optional external calibration step. Reconstruction via the main diagonal is not feasible since the angular differences yield

$$G^{\mathrm{D}}_{ii} = \sin(\phi_i - \phi_i) = 0,$$

regardless of whether the time series is scaled to $[-1, 1]$ with angles in $[0, \pi]$ or to $[0, 1]$ with angles in $[0, \pi/2]$. The method proposed later in Section 4.4 therefore uses off-diagonal angular differences rather than the diagonal.

2.3. Markov Transition Field

2.3.1. Image Construction

Introduced by Wang and Oates in *Imaging Time-Series to Improve Classification and Imputation* [9], the Markov Transition Field (MTF) is probabilistic rather than geometric or trigonometric. Given a time series X, the values are divided into Q quantile bins and each sample is assigned a discrete state $q_t \in \{1, \dots, Q\}$. The first-order Markov transition matrix is estimated as

$$W_{p,q} = P(q_{t+1} = q \mid q_t = p) = \frac{\#\{t : q_t = p,\ q_{t+1} = q\}}{\#\{t : q_t = p\}},$$

with rows normalized to sum to one when the source state is observed.

The MTF expands this $Q \times Q$ matrix back to an $N \times N$ image by assigning each pixel the transition probability associated with the states of its two time indices:

$$M_{a,b} = W_{q_a, q_b}, \qquad a, b = 1, \dots, N. \tag{9}$$

Thus, each pixel stores a transition statistic between quantile states rather than a direct distance or angular relation between amplitudes. This explains why MTF can be robust for classification while being weak for pointwise reconstruction: all samples inside the same quantile share the same reconstructed representative value.

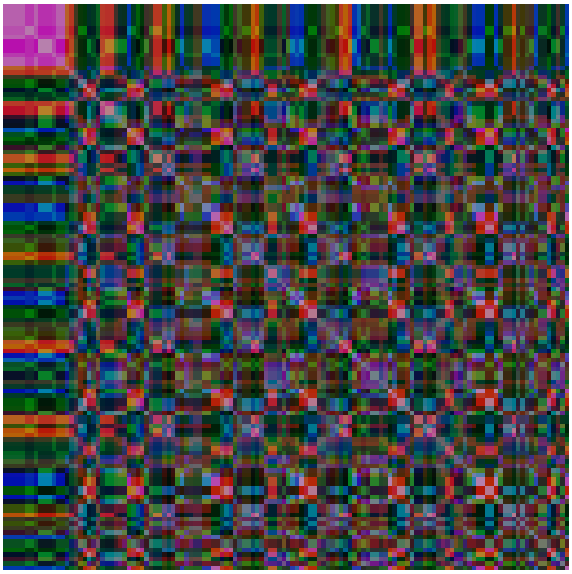


Figure 4: Example of a Markov Transition Field.

2.3.2. Time Series Reconstruction

In the literature, there is no documented method for estimating the original time series from a Markov Transition Field (MTF) matrix. Some articles use Markov Random Fields for image estimation and reconstruction, applying them as image filters to improve reconstructed image quality [12, 13]. However, those methods address image restoration rather than inverse MTF decoding.

3. Problem Formulation

Let $x = (x_1, \dots, x_N)$ denote one axis of a triaxial accelerometer window, with $N = 128$ in all experiments. Each method m first maps the raw window into its own normalised domain $\mathcal{X}_m$ through a per-window, method-specific normaliser $\mathcal{N}_m$ (formalised in Section 4.1), $\tilde{x}^{(m)} = \mathcal{N}_m(x)$; the image construction then acts on that normalised signal,

$$\mathcal{E}_m : \mathcal{X}_m \to \mathbb{R}^{N_m \times N_m},$$

a single-channel matrix per accelerometer axis. In our setting, $N_m = N$ for GASF, GADF, and MTF, while the modified RP representation internally uses adjacent states and therefore has $N_m = N - 1$ before the recovered sequence is re-augmented back to N. The inverse problem studied in this paper is to construct a decoder

$$\mathcal{D}_m : \mathbb{R}^{N_m \times N_m} \to \mathcal{X}_m$$

that returns a *normalised* reconstruction $\hat{\tilde{x}}^{(m)} = \mathcal{D}_m(\mathcal{E}_m(\tilde{x}^{(m)}))$ close to $\tilde{x}^{(m)}$ under pointwise, trend-level, and downstream classification criteria; recovery of the physical-scale window $\hat{x} = \mathcal{N}_m^{-1}(\hat{\tilde{x}}^{(m)}) \in \mathbb{R}^N$ is the optional final step of the shared protocol. A triaxial window is encoded axis by axis; the three single-channel matrices are stacked into one $N_m \times N_m \times 3$ RGB image for storage, and the decoder is applied independently to each channel before the recovered axes are regrouped into a single window.

This formulation separates three sources of difficulty. First, some encodings introduce invariances or many-to-one summaries by construction. In RP, the adjacent-state embedding produces $N-1$ overlapping two-dimensional states that still contain all samples in principle, but the stored pairwise distances or recurrences are invariant to transformations such as translation, rotation, reflection, and, depending on normalization, scale; the decoder then also has to map the recovered $N-1$ state sequence back to N samples. Second, some encodings introduce branch or orientation ambiguities, as in GASF diagonal inversion, GADF sine-difference inversion, and multidimensional scaling for RP. Third, image storage and normalization introduce practical quantisation and rescaling errors. The proposed decoders therefore report their assumptions explicitly: normalization range, required encoder settings or calibration, orientation convention, and whether the reconstruction is exact, approximate, or heuristic.

The rest of the paper follows this formulation. Section 4 describes the decoders, including the proposed GADF multi-pivot, boundary-anchored reconstruction. Section 5 defines the data and metrics, Section 6 reports the complete empirical evidence for all four encodings, and Section 7 interprets the trade-offs among encodings. This organization is intentional: methods may be mathematically defined before all empirical evidence is available, but comparative claims are restricted to the decoders tested under the same protocol.

4. Proposed Reconstruction Methods

This section describes the reconstruction strategy associated with each encoding. Each method is presented as an inverse map with explicit assumptions, so that later experiments can be read as tests of these assumptions rather than as isolated implementation results.

4.1. Shared normalisation, storage, and calibration protocol

All four encoders and decoders operate on a per-window, per-axis normalised signal, and the mapping back to physical units is a single optional external stage shared by every method (the normalisation and optional physical-scale-recovery blocks in Fig. 1). Writing $\mathcal{N}_m$ for the per-window, method-specific normaliser and $\mathcal{E}_m, \mathcal{D}_m$ for the encoder and decoder of method m, the pipeline is

$$\tilde{x} = \mathcal{N}_m(x), \qquad \hat{\tilde{x}} = \mathcal{D}_m\big(\mathcal{E}_m(\tilde{x})\big),$$

where the encoded matrix is stored and re-read as an 8-bit PNG between $\mathcal{E}_m$ and $\mathcal{D}_m$: each matrix M is quantised as $\mathrm{round}(255\, g_m(M))$ into a `uint8`, with the three axes stored as the R/G/B channels of one image; g_m min–max-normalises per image for RP and the two Gramian fields and is the identity on the fixed $[0,1]$ range for MTF, and the decoder applies the corresponding inverse ($[0,255] \to [0,1]$ for MTF, $\to [-1,1]$ for the Gramian fields, and $\to$ the stored per-axis signal range for RP). MTF is therefore the only exactly-invertible store/read pair; for RP and the Gramian fields the read recovers the matrix only up to the per-image normalisation, a discrepancy absorbed by the downstream angular-branch and graph/MDS decoding. **Unless otherwise stated, every decoder returns its reconstruction in the normalised domain** $\hat{\tilde{x}}$; the per-axis minimum and maximum $(x_{\min}, x_{\max})$ recorded at encoding time are used *only* when the normalised series must be mapped back to the physical accelerometer units,

$$\hat{x} = \mathcal{N}_m^{-1}(\hat{\tilde{x}};\, x_{\min}, x_{\max}).$$

Metadata dependence is therefore conditional on the desired output domain: no per-window amplitude metadata are needed for normalised reconstruction, whereas physical-scale reconstruction requires the original scale calibration. The *normalisation range*, by contrast, is part of each encoder–decoder definition and is not interchangeable: the diagonal GASF inverse assumes the $[0,1]$ range (angles in $[0,\pi/2]$), the restricted-branch GADF inverse assumes $[-\alpha,\alpha]$ with $\alpha < 1/\sqrt{2}$, MTF discretises the normalised range into Q quantile bins, and the signed-RP orientation check requires the encoder and decoder to share the same signed-RP normalisation convention. Each subsection below therefore states its normalisation range explicitly, returns a normalised reconstruction, and defers the optional physical-scale step to the common $\mathcal{N}_m^{-1}$ above.

4.2. Recurrence Plot

The Recurrence Plot decoder proposed below is a *new* contribution of this work; the closest precedent in the literature is the unsigned reconstruction of Hirata, Horai, and Aihara [7], which recovers a state ordering through multidimensional scaling but stops there, leaving the geometry defined only up to an orthogonal transformation and never returning to the scalar signal. Our decoder closes that gap by exploiting a property specific to the signed Lu–Tong RP under the L_2 norm: each off-diagonal entry is (up to sign) the Euclidean distance between the two-dimensional adjacent states $s_i = (x_i, x_{i+1})$, so the recovered magnitude matrix is a genuine Euclidean distance matrix between those states. We therefore (i) invert the stored image back to the RP's *own* distance range rather than the signal amplitude range, (ii) recover the full two-dimensional state cloud with classical multidimensional scaling, (iii) fix the residual orthogonal transformation with an overlap constraint between consecutive states and the signed-RP convention, and (iv) read the scalar series directly from the overlapping coordinates. Correcting the inverse normalisation is what makes the recovered matrix a valid distance matrix; the earlier one-dimensional projection reported in preliminary experiments decoded the image with the signal amplitude range, which affinely distorts the distances handed to multidimensional scaling and collapses the two-dimensional geometry. With the correction the encoder–decoder pair is near-invertible on the protocol of this paper (Pearson ≈ 0.99998, mean absolute error ≈ 0.015; Table 6), on par with the Gramian fields.

Multidimensional scaling recovers the two-dimensional state cloud only up to a global orthogonal transformation and translation; the decoder therefore aligns that cloud with the overlap structure of consecutive adjacent states and resolves the residual reflection with the signed-RP convention. Concretely:

- **Step 1 (corrected inverse normalisation):** Recover the signed RP matrix R^{obs} from the decoded image by inverting the PNG quantisation with the RP matrix's *own* stored range $[R_{\min}, R_{\max}]$ — the per-window, per-axis minimum and maximum of the signed RP — and *not* the signal amplitude range. This is the decisive correction: because the encoder min–max-normalises each RP by its own range, inverting with the signal range applies an affine map $R^{\mathrm{obs}} = aR + b$ with $b \neq 0$, which shifts the zero of the distance matrix and destroys its Euclidean structure. The RP range is one extra pair of scalars per window, stored alongside the signal range already recorded for physical-scale recovery.
- **Step 2 (magnitude matrix):** Form the symmetric magnitude matrix $D_{ij} = \frac{1}{2}\big(|R^{\mathrm{obs}}_{ij}| + |R^{\mathrm{obs}}_{ji}|\big)$ with $D_{ii} = 0$; the symmetrisation absorbs the small asymmetries introduced by PNG quantisation. Under the L_2 signed-RP definition, $D_{ij} = \|s_i - s_j\|_2$ is the Euclidean distance between the adjacent states $s_i = (x_i, x_{i+1})$.
- **Step 3 (two-dimensional embedding):** Apply classical (Torgerson) multidimensional scaling to D and retain the two leading positive-eigenvalue coordinates, giving $Y = V_2 \Lambda_2^{1/2} \in \mathbb{R}^{(N-1)\times 2}$. For a noise-free Euclidean D this recovers the state cloud exactly, up to an orthogonal transformation $Q \in O(2)$ and a translation.
- **Step 4 (overlap alignment):** The true states satisfy the overlap identity $s_{i,2} = s_{i+1,1} = x_{i+1}$. We search the orthogonal group for the transform that best enforces it: for each rotation angle θ and reflection parity $\eta \in \{\pm 1\}$ we form $Q_{\theta,\eta}$ and $Z = YQ_{\theta,\eta}$, correct the translation offset $\delta = -\frac{1}{N-2}\sum_i (Z_{i,2} - Z_{i+1,1})$ via $a_i = Z_{i,1} - \delta/2$, $b_i = Z_{i,2} + \delta/2$, and score the orientation by the normalised overlap error $E_{\mathrm{overlap}} = \sum_i (b_i - a_{i+1})^2 \big/ \big(\sum_i (a_i^2 + b_i^2) + \epsilon\big)$.
- **Step 5 (signed orientation selection):** Among the lowest-overlap-error orientations, we regenerate the signed RP of each candidate reconstruction using Equation 4 and keep the one whose signed RP is closest to R^{obs} in Frobenius norm. The signed convention thus resolves the remaining rotation and reflection, generalising the earlier binary sign check that chose only between a reconstruction and its global negation.
- **Step 6 (overlap reconstruction):** From the aligned states $\hat{s}_i = (a_i, b_i)$ the scalar series follows directly from the overlap, recovering all N samples from the $N - 1$ states without any appended constant:

$$\hat{x}_1 = a_1, \qquad \hat{x}_i = \tfrac{1}{2}\big(b_{i-1} + a_i\big) \;\; (2 \le i \le N-1), \qquad \hat{x}_N = b_{N-1}.$$

  Averaging the two estimates of each shared sample also attenuates residual multidimensional-scaling and quantisation noise. When physical accelerometer units are required, the normalised amplitudes are mapped onto $[x_{\min}, x_{\max}]$ through the shared calibration $\mathcal{N}_m^{-1}$ of Section 4.1.

With the corrected inverse normalisation the recovered D is, up to quantisation, a valid two-dimensional Euclidean distance matrix: multidimensional scaling explains essentially all of its variance in two dimensions and the overlap reconstruction returns the signal almost exactly (Table 6). The inverse-normalisation correction and the signed orientation selection are the two decisive ingredients; without the former the distances are affinely distorted and the two-dimensional geometry collapses, and without the latter the reflection remains unresolved.

A more rigorous alternative, which we identify here as the principled version of this decoder but do not evaluate in the present experiments, would retain *both* positive multidimensional-scaling dimensions $\hat{s}_i = (\hat{s}_{i,1}, \hat{s}_{i,2})$ instead of collapsing each state onto its largest-eigenvalue axis. Since the two-dimensional embedding is recovered only up to a global orthogonal transformation (rotation and reflection) together with a translation and scale, one would estimate

the orthogonal transformation that enforces the overlap consistency of consecutive states, $\hat{s}_{i,2} \approx \hat{s}_{i+1,1}$, for example by minimising $\sum_i \left(\hat{s}_{i,2} - \hat{s}_{i+1,1}\right)^2$ over the orthogonal group. The signal would then follow directly from the overlap rather than from an appended constant:

$$\hat{x}_1 = \hat{s}_{1,1}, \qquad \hat{x}_{i+1} \approx \tfrac{1}{2}\big(\hat{s}_{i,2} + \hat{s}_{i+1,1}\big), \qquad \hat{x}_N = \hat{s}_{N-1,2},$$

which recovers all N samples from the $N-1$ states without a heuristic length correction. The signed-RP convention of Equation 4 would then be used to select, among the orthogonal transformations compatible with the encoded image, the one matching its sign structure, rather than to choose between the two signs of a single projected coordinate. We leave this two-dimensional variant as future work; the decoder evaluated in this paper is the one-dimensional projection described above.

4.3. Gramian Angular Summation Field Reconstruction

Unlike the other three decoders in this section, the GASF inverse is *not* a new contribution of this work: it follows the diagonal-inverse strategy used by Huamin et al. [11] in the 2-D representation-based denoising autoencoder line of work. We re-derive it here in the notation of the rest of the paper and re-implement it inside the same pipeline used by the other three decoders, so the GASF numbers reported later can be read as a tightly controlled prior-art baseline against which the three novel decoders (RP, GADF, MTF) are compared. The contribution of this subsection is therefore the explicit normalisation and quantisation handling that make the inverse stable when the GASF matrix is read back from an 8-bit PNG, not the inverse map itself.

The GASF decoder is the most direct inverse considered in the completed experiments because its main diagonal contains a sample-wise angular constraint. Under the branch used in this work, the encoded values are first interpreted through

$$G^{\mathrm{S}}_{ii} = \cos(2\phi_i),$$

and the angle is recovered as

$$\hat{\phi}_i = \frac{1}{2}\arccos\left(G^{\mathrm{S}}_{ii}\right).$$

The normalized signal is then obtained by

$$\hat{\tilde{x}}_i = \cos(\hat{\phi}_i).$$

This rule is exact only when the normalization and angular branch are consistent with the encoder. If the full interval $[-1, 1]$ is used without additional sign information, the diagonal determines $|\tilde{x}_i|$ rather than $\tilde{x}_i$ itself. Therefore, the high numerical accuracy reported for GASF should be read as the result of a controlled normalization protocol, not as an unconditional inverse for every possible GAF image.

The recovered $\hat{\tilde{x}}_i = \cos(\hat{\phi}_i)$ is the normalised reconstruction; when physical accelerometer units are wanted, it is passed through the common calibration $\mathcal{N}_m^{-1}$ of Section 4.1. The normalised shape is thus recovered from the diagonal alone, and the recorded per-window min–max is needed only for the optional physical-scale step — which is also why a generated GASF image would require plausible scale metadata only if physical units, rather than normalised shape, are required.

4.4. Restricted-Branch Gramian Angular Difference Field Reconstruction

We now describe a restricted-branch GADF decoder. Wang and Oates note that GADF can admit an accurate inverse map when the data are rescaled to $[0, 1]$ [9], but their reconstruction discussion operationalises the inverse through the GASF diagonal and does not specify a GADF decoding algorithm. In particular, GADF stores $\sin(\phi_i - \phi_j)$, so an operational inverse must choose an arcsin branch, fix the otherwise free global angular offset, keep a sign convention fixed, and map the recovered angular trajectory back to normalised amplitudes, with physical-unit recovery left as an optional external calibration step. The contribution here is therefore not the abstract observation that GADF may be invertible under a restricted angular range, but a fully specified operational decoder: a closed-form multi-pivot, boundary-anchored reconstruction with an explicit branch condition, an image-derived angular anchor, a fixed sign convention, optional min–max calibration for physical-scale recovery, and empirical evaluation under the same protocol as RP, MTF, and GASF. It should not be read as a generic inverse for every standard GADF image on the full $[-1, 1]$ range. The novelty lies in two coupled design choices: (i) we shrink the normalised amplitude range to $[-\alpha, \alpha]$ with $\alpha = 0.7$ so that *every* pairwise angular difference is forced inside the principal branch of arcsin, which is what turns the inverse from an iterative optimisation into a one-pass closed-form computation, and (ii) we anchor the otherwise-relative angular trajectory by reading the angular boundary directly off the extreme row of the GADF image (rather than from an external time-domain index), so that the normalised trajectory is recovered from the image alone — the stored $(x_{\min}, x_{\max})$ being needed only for optional physical-scale recovery.

GADF requires a reconstruction strategy different from the GASF diagonal inverse. The diagonal of G^{D} is identically zero, but the full matrix stores relative angular information through $G^{\mathrm{D}}_{ij} = \sin(\phi_i - \phi_j)$. The proposed decoder uses this off-diagonal structure and turns a row of the GADF matrix into a set of recoverable angular differences.

4.4.1. Restricted scaling for branch uniqueness

The general GADF inverse is not unique without additional assumptions because $\sin(\theta) = \sin(\pi - \theta)$. The key design decision is therefore to restrict the normalized amplitude range before encoding. This sacrifices part of the nominal $[-1, 1]$ GAF range, but it keeps all pairwise angular differences inside the principal branch of arcsin, which is what makes a closed-form decoder possible. The input series is scaled as

$$\bar{x}_t = 1.4\frac{x_t - x_{\min}}{x_{\max} - x_{\min}} - 0.7, \qquad \bar{x}_t \in [-\alpha, \alpha], \quad \alpha = 0.7.$$

The corresponding angles satisfy

$$\phi_t = \arccos(\bar{x}_t) \in [\arccos(\alpha), \arccos(-\alpha)] \approx [0.795, 2.346].$$

The value $\alpha = 0.7$ is not an arbitrary hyperparameter. It is chosen as a near-maximal symmetric amplitude range that preserves the injectivity of the inverse-sine step. For the closed-form GADF inverse to be unique, every angular difference must remain inside the principal branch of arcsin:

$$|\phi_i - \phi_j| < \frac{\pi}{2}.$$

The worst-case angular spread is

$$\arccos(-\alpha) - \arccos(\alpha).$$

Using $\arccos(-\alpha) = \pi - \arccos(\alpha)$, the uniqueness condition becomes

$$\pi - 2\arccos(\alpha) < \frac{\pi}{2},$$

or equivalently

$$\alpha < \cos\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} \approx 0.7071.$$

We therefore set $\alpha = 0.7$, close to the largest admissible symmetric range while leaving a small numerical margin:

$$|\phi_i - \phi_j| \leq \arccos(-0.7) - \arccos(0.7) \approx 1.551 < \frac{\pi}{2}.$$

This keeps every angular difference within the principal branch of arcsin. Under this restriction,

$$\arcsin\left(G^{\mathrm{D}}_{ij}\right) = \phi_i - \phi_j.$$

Without the restriction $\alpha < 1/\sqrt{2}$, the same value $\sin(\phi_i - \phi_j)$ could correspond to multiple angular-difference configurations, and the GADF matrix would no longer determine the angular differences uniquely. The branch restriction gives a single angular reconstruction under the controlled protocol, once the absolute angular anchor is also fixed (the min–max metadata is not needed for this normalised reconstruction, only for the optional physical-scale recovery). The reliability check of Section 6.5 confirms at $\alpha = 0.7$ that the floating-point inverse is exact, separates it from 8-bit PNG quantisation error, and finds a 0% anchor-misidentification rate on WISDM.

4.4.2. Relative angular reconstruction from a pivot row

For a pivot row p, the GADF entries provide

$$G^{\mathrm{D}}_{pj} = \sin(\phi_p - \phi_j),$$

and, by the branch condition above,

$$\phi_j = \phi_p - \arcsin\left(G^{\mathrm{D}}_{pj}\right), \qquad j = 1, \ldots, N.$$

Thus a pivot row determines the entire angular trajectory only up to a global offset; the absolute offset is fixed separately by the boundary (extreme-mean) anchor described below. The pivot is chosen from the image itself rather

than from an external time-domain index, because a stored or generated GADF image may not preserve the original contextual information. We select the pivot row as

$$p = \arg\min_i \left| \frac{1}{N} \sum_{j=1}^{N} G_{ij}^{\mathrm{D}} \right|,$$

which chooses the row whose average signed angular difference is closest to zero. Because the row mean equals $R\sin(\phi_i - \delta)$ (derived below), this selects rows whose angle ϕ_i is close to the circular-mean angle δ of the window — *not*, in general, to $\pi/2$: if the amplitudes concentrate near one extreme, δ can drift toward that boundary even though per-window normalisation guarantees both extremes are present. In practice the decoder does not rely on a single pivot: it averages the reconstruction over the $n_p = 5$ most-central rows (the five smallest |row mean|), which cancels most of the per-row 8-bit quantisation noise (Section 6.5 quantifies the gain). Selecting central pivots *tends* to keep the recovered differences $G_{pj}^{\mathrm{D}} = \sin(\phi_p - \phi_j)$ small, so most of the arcsin arguments the decoder reads fall in its near-linear regime where 8-bit quantisation is least amplified. This is an *empirical* tendency rather than a guarantee: on WISDM the selected pivots sit close to the angular centre (mean $|\phi_p - \frac{\pi}{2}| \approx 0.13$ rad) and the pivot-row entries are small on average (mean $|G_{pj}^{\mathrm{D}}| = 0.22$, 99th percentile 0.76), but the worst case can still approach the branch boundary (max ≈ 0.99) for windows whose amplitude distribution is strongly skewed within the admissible interval.

4.4.3. Absolute angular anchor

The pivot row reconstructs angles relative to ϕ_p, so one absolute angular anchor is still required — but this anchor is supplied by the *known* normalisation range, not by the stored min–max. Because the normalisation is performed per window, the encoded normalised signal necessarily reaches the two boundary values $+\alpha$ and $-\alpha$ unless the window is constant, so the extreme samples sit at fixed, *a priori* known angular boundaries:

$$\phi_{\max} = \arccos(\alpha), \qquad \phi_{\min} = \arccos(-\alpha).$$

Operationally, the boundary row is identified from the image as the extreme row

$$e = \arg\max_i \left| \frac{1}{N} \sum_{j=1}^{N} G_{ij}^{\mathrm{D}} \right|.$$

The sign of the mean row indicates which boundary is represented: a positive mean corresponds to the large-angle boundary $\phi_e = \arccos(-\alpha) = \phi_{\min}$, whereas a negative mean corresponds to the small-angle boundary $\phi_e = \arccos(\alpha) = \phi_{\max}$.

This identification is exact in the noise-free restricted regime, because the row mean is a strictly monotonic function of the row angle. Writing $\bar{C} = \frac{1}{N}\sum_j \cos\phi_j$ and $\bar{S} = \frac{1}{N}\sum_j \sin\phi_j$,

$$\frac{1}{N}\sum_{j=1}^{N} G_{ij}^{\mathrm{D}} = \bar{C}\sin\phi_i - \bar{S}\cos\phi_i = R\,\sin(\phi_i - \delta), \qquad R = \sqrt{\bar{C}^2 + \bar{S}^2}, \quad \delta = \operatorname{atan2}(\bar{S}, \bar{C}),$$

whose derivative $R\cos(\phi_i - \delta)$ is strictly positive over the whole admissible range. Indeed every $\phi_j \in [\arccos\alpha, \arccos(-\alpha)]$, and δ is the argument of a convex combination of the unit vectors $(\cos\phi_j, \sin\phi_j)$, so δ lies in that same interval and $\phi_i - \delta \in [-(\arccos(-\alpha) - \arccos\alpha), \arccos(-\alpha) - \arccos\alpha]$. For $\alpha = 0.7$ this spread is $\approx 1.551 < \pi/2$ — the same margin that keeps the arcsin branch unique — hence $\cos(\phi_i - \delta) > 0$ throughout and the row mean is strictly increasing in ϕ_i. Its extreme value is therefore attained exactly at the boundary sample ($\pm\alpha$) that the per-window normalization is guaranteed to produce, and the sign of that extreme selects the correct boundary. The $\alpha = 0.7$ restriction thus secures branch uniqueness and unambiguous anchor identification at once.

Once a boundary row is identified, it fixes the otherwise-free global offset through a *single* entry per pivot: for each central pivot p,

$$\phi_p = \phi_e - \arcsin\left(G_{ep}^{\mathrm{D}}\right),$$

after which the full angular trajectory follows from that pivot row — not from the extreme row — through $\phi_j = \phi_p - \arcsin(G_{pj}^{\mathrm{D}})$ of the previous subsection. The decoder does not trust a single pivot: it averages the recovered *angles* over the set P of n_p central rows and only then maps back to amplitude,

$$\hat{\phi}_j^{(p)} = \phi_e - \arcsin\big(G_{ep}^{\mathrm{D}}\big) - \arcsin\big(G_{pj}^{\mathrm{D}}\big), \qquad \hat{\phi}_j = \frac{1}{n_p}\sum_{p\in P} \hat{\phi}_j^{(p)}, \qquad \hat{\tilde{x}}_j = \cos\Big(\hat{\phi}_j\Big), \quad j = 1, \ldots, N.$$

Averaging the *angles* — not the reconstructed amplitudes — is what the implementation does, and it is what lowers the reconstruction error as n_p grows (Section 6.5). The reconstruction is unique within the controlled protocol: the

central pivot rows supply the trajectory shape, while the boundary value imposed by the per-window normalisation — read through the anchor scalar G^{D}_{ep} (empirically ≈ 0.78 on WISDM: moderate, though, unlike the 0.70 of an exactly-central pivot, not pinned away from the branch edge) — fixes the global offset. If the stored matrix uses the transposed GADF convention $\sin(\phi_j - \phi_i)$, every arcsin sign above flips; the implementation must therefore keep the convention in Equation 8 fixed.

4.4.4. Normalised output and limitations

The GADF image, the fixed value of α, and the encoding sign convention are sufficient to recover the normalised trajectory $\hat{\tilde{x}}_j = \cos(\phi_j) \in [-\alpha, \alpha]$ directly from the image; the per-window minimum and maximum are *not* required for this step. The calibration pair $(x_{\min}, x_{\max})$ enters only through the common physical-scale map $\mathcal{N}_m^{-1}$ of Section 4.1, i.e. only when the reconstructed series must be expressed in the original accelerometer units; metadata dependence is thus conditional on the output domain. The method is closed-form but not linear-time overall: selecting the anchor and pivot rows requires the row-mean statistics over the full $N \times N$ input matrix, so the decoder has $O(N^2)$ total cost — dominated by those row statistics (equivalently, by reading the N^2 stored pixels) — followed by an $O(N)$ closed-form angular reconstruction from each selected row once the anchor is fixed. It is also not universally invertible: it relies on the restricted scaling interval $[-\alpha, \alpha]$ with $\alpha < 1/\sqrt{2}$, a consistent GADF sign convention, and the presence or reliable identification of an angular boundary. Stating these requirements is essential for the paper's scope: the decoder is a controlled inverse map, not a guarantee that every visually plausible GADF image corresponds to a valid physical signal. A quantisation caveat bounds the *worst case* rather than the operating point: the entry between the two extreme samples reaches $|G^{\mathrm{D}}| = \sin\big(\arccos(-\alpha) - \arccos\alpha\big) \approx 0.9998$, where arcsin has derivative ≈ 50, so one 8-bit PNG step ($\approx 2/255$ in G^{D}, since the stored value spans $\approx [-1, 1]$ over the 256 levels) there maps to an angular error of order 0.4 rad. The multi-pivot decoder, however, mostly avoids that entry: it reconstructs each trajectory from central rows whose arcsin arguments are small in typical windows (on WISDM, mean $|G^{\mathrm{D}}_{pj}| = 0.22$, 99th percentile 0.76) and reads the extreme row through one scalar G^{D}_{ep} per pivot (mean ≈ 0.78). For almost all windows the steep near-boundary region is therefore off the reconstruction path; but because the central-pivot criterion only *tends* to place pivots near the angular centre, individual entries can still approach the boundary (max ≈ 0.99 on WISDM), so this is an empirical rather than a guaranteed property. It nonetheless explains why the measured PNG error sits far below the worst-case bound (Section 6.5: floating-point MAE $< 10^{-6}$, 8-bit PNG MAE 0.0158 at $\alpha = 0.7$ with $n_p = 5$). The decoder is thus exact in real arithmetic and, empirically, keeps the quantisation-sensitive boundary off its typical reconstruction path; image-perturbation robustness is left to future work. When these assumptions fail, a more general nonlinear least-squares reconstruction can be used:

$$\min_{\phi} \sum_{i<j} w_{ij} \left(\sin(\phi_i - \phi_j) - G^{\mathrm{D}}_{ij}\right)^2,$$

with one angle fixed to remove the global phase ambiguity and with smoothness or locality weights w_{ij} for robustness.

4.5. Markov Transition Field Reconstruction

The MTF decoder described below is, to the best of our knowledge, not preceded by a documented explicit MTF-to-time-series decoder in the literature reviewed here. The decoder is not bijective: a Markov chain induces an equivalence class of time series with the same transition statistics, and no algorithm can distinguish elements of that class from the MTF image alone. What follows recovers a representative of that class by clustering the MTF rows into quantile bins and fixing the bin ordering through the Fiedler vector of the recovered transition graph. This spectral ordering step is motivated by the seriation literature, where the Fiedler vector can recover a latent linear order under appropriate structural conditions [16]; in our setting, the exact-order statement should therefore be read as an ideal birth–death-chain case, not as a universal guarantee for arbitrary MTF images.

The reconstruction proceeds in four steps:

- **Step 1: Bin recovery.** Cluster the T rows of the MTF matrix $M \in \mathbb{R}^{T \times T}$ into Q groups by farthest-point-initialised k-means on the row-normalised image. Under a clean MTF every sample in the same quantile bin has the same row pattern, so this recovers the bin label sequence $\{q_1, \ldots, q_T\}$ up to a bin permutation.
- **Step 2: Bin ordering via the Fiedler vector.** Estimate the bin transition matrix W as the average of MTF entries inside every (p, q) block, symmetrise it as $S = \frac{1}{2}(W + W^\top)$, and order the bins by the Fiedler vector of the graph Laplacian $L = D - S$ (the eigenvector corresponding to the second-smallest eigenvalue). For an ideal birth–death Markov chain on Q ordered states this recovers the linear amplitude order; for general chains it is a spectral seriation heuristic whose quality depends on how strongly the recovered transition graph reflects an underlying one-dimensional amplitude order.

- **Step 3: Amplitude assignment.** For every time index i belonging to sorted bin G_k, assign the midpoint of the corresponding quantile interval as the reconstructed normalised value:

$$\hat{x}_i = \frac{k - \frac{1}{2}}{Q}, \qquad k = 1, \ldots, Q.$$

- **Step 4: Normalised output.** Steps 1–3 already return the reconstruction in the normalised domain (bin-midpoint values in $[0, 1]$); recovery of the physical scale, when required, follows the common calibration $\mathcal{N}_m^{-1}$ of Section 4.1.

The procedure does not verify the global sign of the series, because the MTF representation emphasises transition structure rather than signed amplitude order; a single sign-bit per axis (not stored here) would resolve the remaining mirror ambiguity. We use $Q = 4$ bins throughout the reported experiments. The choice is defended both by a noise-of-estimator argument – with $T = 128$ samples and Q bins there are $\sim (T-1)/Q^2$ transitions per cell of the empirical row-stochastic matrix W, so $Q = 4$ admits ~ 8 transitions per cell while $Q = 8$ drops to ~ 2 and many cells of W become degenerate at 0 or 1 – and empirically by the bin-count ablation in Section 6.7, where $Q = 4$ is the value that simultaneously minimises the pointwise MAE, maximises the pointwise Pearson correlation, and maximises the amplitude-sensitive KNN accuracy (CNN nearly unchanged between $Q = 2$ and $Q = 4$). The remaining classifiers (SVM, Random Forest, LSTM, GRU) vary by at most 0.012 accuracy across the entire sweep, so the recommendation of $Q = 4$ is dominated by the amplitude-sensitive subset rather than overridden by the others.

5. Experimental Protocol

The experimental protocol is designed to answer three separate questions: whether the signal can be reconstructed point-by-point, whether its global trend is preserved, and whether the reconstructed window still supports activity recognition. These questions are intentionally separated because a representation may be poor for sample-level reconstruction while still retaining enough coarse structure for classification. This section fixes the dataset, segmentation, qualitative checks, metrics, and classifier protocol before the results are reported.

5.1. Dataset and Segmentation

The dataset used is the *WISDM Smartphone and Smartwatch Activity and Biometrics Dataset* [14]. It contains data from 51 subjects, with accelerometer and gyroscope measurements collected from a smartwatch worn on the dominant hand and a mobile device kept in the pocket. The dataset includes 18 activities, each recorded for 3 minutes. We use the smartwatch accelerometer stream, which is sampled at 20 Hz (i.e., one sample every 50 ms).

For our experiments, we selected five common activities with nearly balanced sample counts. This subset keeps the classification task interpretable while avoiding a strongly imbalanced evaluation in which reconstruction effects could be masked by class-frequency effects. The chosen activities are:

Table 3: Distribution of activities in the dataset.

Activity	Raw sensor samples	Percentage
Standing	216,529	20.6%
Sitting	213,018	20.2%
Walking	210,495	20.0%
Stairs	207,312	19.7%
Jogging	205,787	19.5%

We segmented the data into fixed non-overlapping windows (FNOW) of length $N = 128$, corresponding to 6.4 seconds at 20 Hz. The window length $N = 128$ (6.4 s) is a common choice for convolutional and recurrent backbones in human-activity recognition, and as a power of two it keeps the GASF, GADF, and MTF images at 128×128 pixels. For the modified RP formulation, adjacent states are still written as $s_j = (x_j, x_{j+1})$, so a length-N window yields an $(N-1) \times (N-1) = 127 \times 127$ image; the reconstruction pipeline then augments the recovered $N - 1$ samples with the per-window mean (step 6.1 of the Hirata–Horai–Aihara adaptation) so that the final signal matches the 128-sample window length used by the other encodings. With $N_{\text{bins}} = 4$ the forward MTF binning is by amplitude value-quantiles, so each of the four bins contains *approximately* $N/Q = 32$ time indices — exactly 32 only in the absence of values tied at a quantile edge.

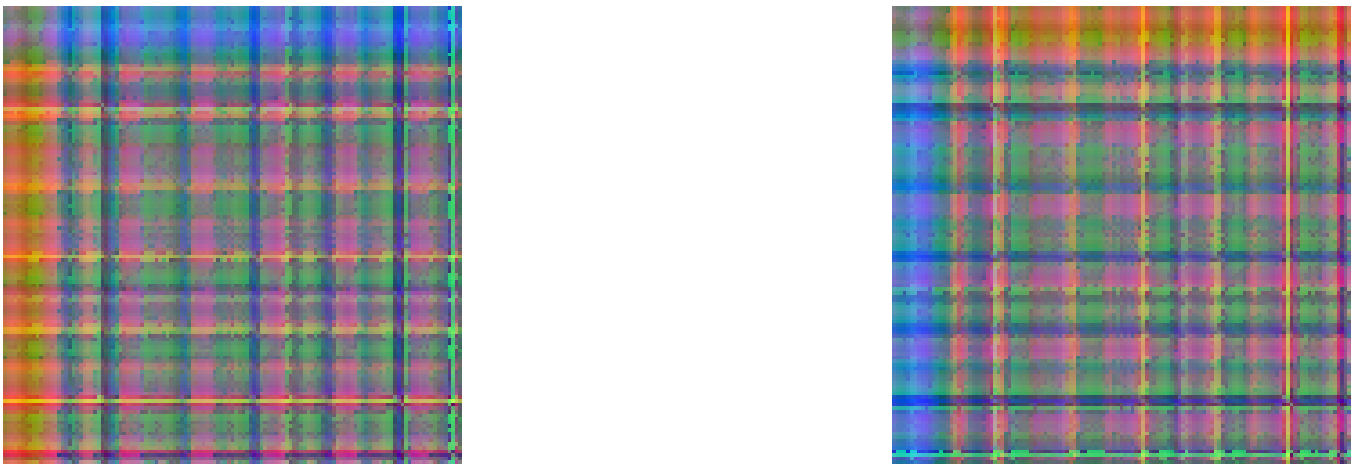
By decomposing the X, Y, and Z dimensions into individual time series of size 128, we obtain 8 172 windows per channel, resulting in a total of $8\,172 \times 3 = 24\,516$ one-dimensional reconstructions for the reconstruction metrics. For classification, however, the experimental unit is the full triaxial window: the three reconstructed axes are grouped

back into a single sample, yielding 8 172 labeled windows. The stratified window-level protocol assigns these to three folds of ∼2 724 test windows each, so the pooled paired-sample size used by the inferential tests in Section 5.3.1 is $N = 8\,172$.

Windowing and grouping details. Windowing is performed *independently within each (subject, activity) segment*: inside one contiguous recording of a given activity by a given subject we take non-overlapping blocks of $N = 128$ samples (stride 128) and discard the trailing remainder shorter than 128. No window therefore crosses a subject or an activity boundary, and every window carries an unambiguous single label and subject id. Dropping the per-segment remainders is what reduces the naive global-slide count to the 8 172 windows used here, and it removes the boundary-straddling windows that a global slide would otherwise mislabel. We evaluate two complementary 3-fold protocols. The stratified window-level split uses a grouping key that is effectively unique per window, so it reduces to a class-stratified partition at the window level: although no single window ever appears in both train and test, different — and often temporally adjacent — windows from the same subject and activity are distributed across the two sets. It therefore measures robustness to reconstruction-induced distribution shift *within* the observed population rather than trial- or subject-level generalisation, and its accuracies should be read as an optimistic within-distribution bound. The grouped subject-out split instead uses the subject id as the grouping key, so every window of a given subject falls entirely in one fold and no test subject is seen during training; this is the conservative protocol for generalisation to unseen individuals.

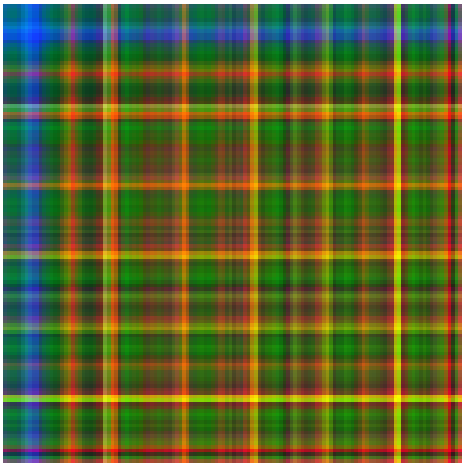
5.2. Encoding Sensitivity to Signal Inversion

Before evaluating reconstructions, we first test whether each encoding visually distinguishes a time series from its sign-inverted counterpart. This check is important because an encoding that maps both signals to the same image cannot by itself resolve orientation during reconstruction. The same WISDM window (subject 1600, first walking window, $N = 128$) is used for every encoder so the only difference across panels is the encoder itself; the inverted variant negates all three accelerometer channels.

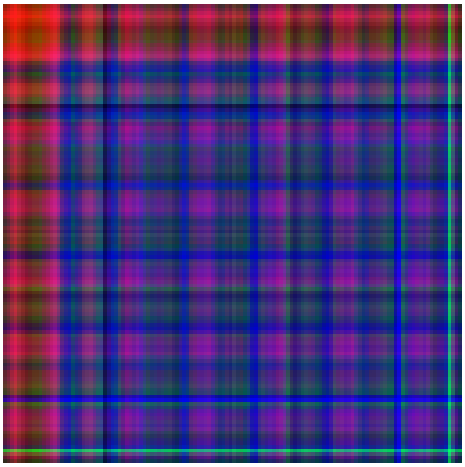


(a) Original (b) Inverted

Figure 5: Recurrence Plot (RP) images: original vs. inverted.

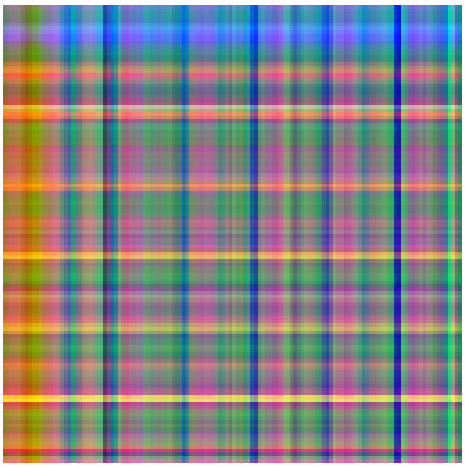


(a) Original

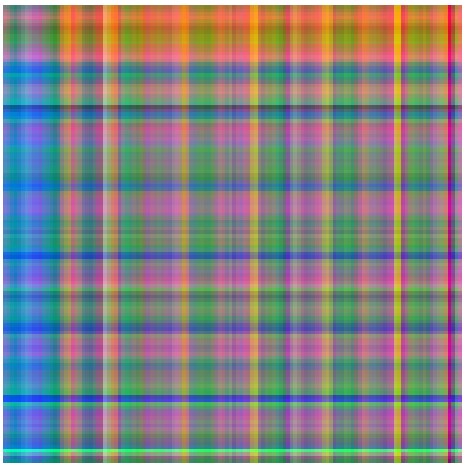


(b) Inverted

Figure 6: Gramian Angular Summation Field (GASF) images: original vs. inverted.

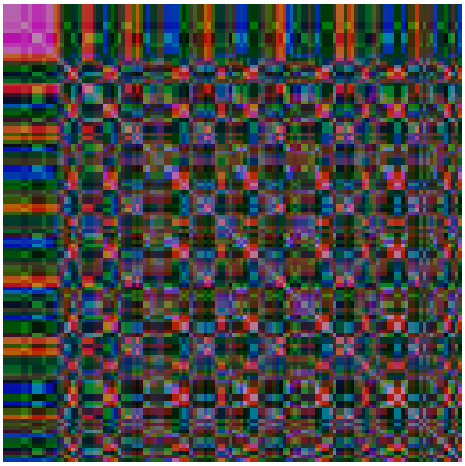


(a) Original

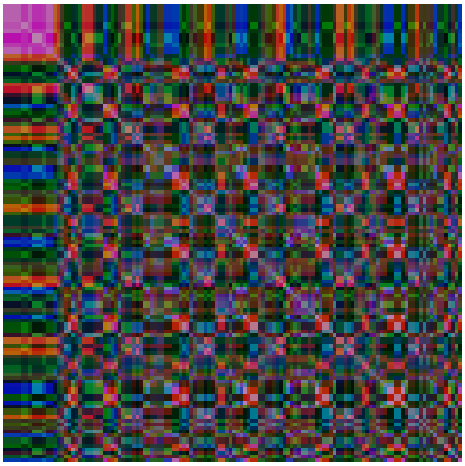


(b) Inverted

Figure 7: Gramian Angular Difference Field (GADF) images: original vs. inverted.



(a) Original



(b) Inverted

Figure 8: Markov Transition Field (MTF) images: original vs. inverted.

As illustrated in Figures 5 (RP) and 6 (GASF), the original and inverted time series produce distinctly different images. The same holds for the Gramian Angular Difference Field (Figure 7): negating the input flips the sign of $\sin(\phi_i - \phi_j)$, which after normalisation appears as a contrast inversion of the GADF image. Conversely, as shown in Figure 8, the Markov Transition Field (MTF) yields identical representations for both the original and inverted series, failing to capture their inversion-specific differences. This invariance is *not* because the transition statistics are themselves unchanged: a global sign flip reverses the quantile ranks, $q \mapsto Q{+}1{-}q$, which reverses both the bin

ordering and the rows and columns of the transition matrix, $W_{p,q} \mapsto W_{Q+1-p,\,Q+1-q}$. The two reversals cancel at the pixel level — $M_{a,b} = W_{q_a,q_b}$ maps to $W_{Q+1-(Q+1-q_a),\,Q+1-(Q+1-q_b)} = W_{q_a,q_b}$ — so the image is left unchanged. The MTF therefore discards the global sign by construction, consistent with the missing sign bit noted in Section 4.5.

5.3. Evaluation Design

We evaluated the four reconstruction pipelines with a three-tier design. The first tier measures sample-level fidelity, the second tier measures trend-level similarity, and the third tier measures downstream discriminative utility. This structure is important because the three questions are not equivalent: a method can have large pointwise errors but still preserve enough temporal structure for a classifier, or it can reconstruct amplitudes accurately while losing information that matters for a specific task. Prior work on regression error metrics and time-series similarity supports using complementary measures when each metric is tied to a specific interpretation [15, 17].

In the first tier, MAE, MRE, MSE, and Pearson correlation are used to compare the original and reconstructed samples directly. MAE gives an interpretable average absolute deviation, MSE penalizes large local failures, MRE normalizes error by signal magnitude, and Pearson correlation measures whether the temporal pattern is linearly aligned even when amplitudes differ. No single metric is sufficient, so these values are interpreted together rather than collapsed into one score.

In the second tier, each min–max normalized window is summarized by a global cubic polynomial. The coefficient vector is (a_3, a_2, a_1, a_0) in the standard power basis, not a piecewise cubic-spline representation. This choice intentionally discards local sample-level detail and tests whether the reconstructed signal preserves broad trend shape. Euclidean and Manhattan distances measure coefficient magnitude differences, cosine similarity measures directional agreement, and Mahalanobis distance accounts for the covariance structure of the original coefficient vectors [18].

In the third tier, we measure original-trained classifier robustness to reconstructed inputs. Each classifier is trained once on the original triaxial windows of the training split and then evaluated, without retraining, on the original and reconstructed versions of the same held-out windows. Classification is included because reconstructed signals should retain task-relevant information, but this experiment measures distribution shift relative to a raw-signal classifier; it is not a direct augmentation experiment. Table 4 separates the scenario reported in this paper from the additional train/test regimes that would be needed to claim augmentation utility. The per-window encoder metadata reused by the decoders (per-axis min–max, the MTF quantile interval, the sign conventions, and α) is part of this controlled inverse problem and must not be interpreted as an augmentation setting, where such metadata would be unavailable.

We state the evaluation domain explicitly, because it differs by tier. The pointwise-error tier (tier one), the visual overlays, and the classifier tier (tier three) are all computed *after* the optional physical-scale map $\mathcal{N}_m^{-1}$ has been applied: reconstructions are expressed in the original accelerometer units, the classifiers are trained and evaluated on those units, and the reported MAE/MRE/MSE are therefore directly comparable to the native amplitude range of the signal. The polynomial-trend tier (tier two) is the sole exception: its cubic coefficients are fitted on min–max-normalised windows so that the trend comparison is scale-invariant, hence its distances live in the normalised domain by construction. This split is deliberate — pointwise magnitude only means something in physical units, whereas trend shape is most fairly compared after removing per-window scale.

Table 4: Classification train/test regimes and the claim each one supports. The reported classifier tables use scenario B; scenarios C and D are listed for completeness and are *not* evaluated in this work, so no data-augmentation benefit is claimed.

Scenario	Train	Test	Main question measured
A	Original	Original	Raw-signal classifier baseline
B	Original	Reconstructed	Robustness to reconstruction-induced distribution shift
C	Reconstructed	Reconstructed	Utility inside the reconstructed domain
D	Original + Reconstructed	Original	Data-augmentation benefit for raw-signal classification

5.3.1. Statistical Significance Protocol

Every comparison reported in this paper is paired. For each stratified window-level fold, the four reconstructions and the original signal are evaluated on the same ∼2 724 test windows, and each classifier is fit once per fold and reused across the five sources within that fold. Per-fold paired correctness vectors are then concatenated across folds 0, 1 and 2, yielding a pooled paired sample of $N = 8\,172$ windows that drives all the inferential tests. Pooling is admissible here as an approximate paired comparison because each window is a single paired observation – it never appears both in training and in test within the same fold, and the test set of a fold never contains a window from any other fold's test set. One caveat remains: because each fold is scored by its own separately-trained model, the pooled 2×2 table aggregates discordant pairs produced under three different decision rules, so the homogeneity assumption behind a

single McNemar table holds only approximately and the resulting p-values should be read as mildly anti-conservative. A stratified Cochran–Mantel–Haenszel or per-fold McNemar would be the more conservative alternative; since all but a handful of the reported comparisons reach $p_{\text{Holm}} < 10^{-9}$, this caveat materially affects only the near-threshold cases we flag as such — chiefly the RP-vs-MTF gaps that shrink once the tuned $\rho = 0.60$ lifts RP, such as GRU under the window-level split ($p_{\text{Holm}} = 1.3 \times 10^{-3}$) and SVM, which becomes a non-significant tie ($p_{\text{Holm}} = 1.0$). A second concern is that the pooled window sample is not fully independent — many test windows share a subject, so window-level p-values could in principle be inflated by intra-subject pseudoreplication. We therefore repeat the pointwise comparison at the subject level, collapsing each subject to a single mean-MAE value so that the $N = 51$ subjects, rather than the 8 172 windows, are the independent units (reported in Section 7.3).

For *accuracy reporting*, every accuracy cell in the per-classifier tables (Tables 20, 21, 22, 23, 24, and 25) is the mean of the three fold accuracies and the dispersion is reported as the standard deviation across folds. Fold-level dispersion is a more honest measure of generalisation than a within-fold bootstrap, which only captures sampling noise inside a single split.

For *paired classifier comparisons* we use the continuity-corrected McNemar test on the pooled discordant pairs [20] – the standard test when two classifiers are evaluated on the same instances – and we distinguish two test families:

- **$\mathbf{F_{M1}}$: Original vs. reconstruction.** For every one of the 4 reconstructions (RP, MTF, GASF, GADF) and every one of the 6 classifiers we test whether the paired correctness vector differs from the original-signal baseline. This produces $4 \times 6 = 24$ paired tests. These McNemar tests are degradation/difference tests, not formal equivalence tests; absence of significance is therefore reported as "no detected degradation" rather than proof of equivalence.
- **$\mathbf{F_{M2}}$: encoder vs. encoder at fixed classifier.** For every pair of reconstructions and every classifier we ask which encoder yields the higher accuracy under that classifier. There are $\binom{4}{2} = 6$ encoder pairs and 6 classifiers, hence $6 \times 6 = 36$ paired tests.

For *paired pointwise-error comparisons* between encoders we use the Wilcoxon signed-rank test on per-window MAE [21]. This test compares two reconstructions on the same windows and does not involve a classifier, so the family contains only $\binom{4}{2} = 6$ tests ($\mathrm{F_W}$). Alongside each Wilcoxon p-value we report a non-parametric effect size $r = Z/\sqrt{N}$, where $|r| \geq 0.5$ is conventionally read as a large effect [22]. The specific GASF-vs-GADF fidelity gap is assessed through this same paired per-window MAE test and, to guard against intra-subject pseudoreplication, through a subject-clustered Wilcoxon that collapses each subject to a single mean-MAE value ($N = 51$ independent units).

Because the three families $\mathrm{F_{M1}}$, $\mathrm{F_{M2}}$ and $\mathrm{F_W}$ contain 24, 36 and 6 simultaneous tests respectively, raw p-values within each family are corrected with the Holm–Bonferroni step-down procedure [23] at a family-wise error rate $\alpha = 0.05$. Throughout the paper we report the Holm-adjusted value p_{Holm} and call a comparison significant only when $p_{\text{Holm}} < \alpha$ inside its own family; the unadjusted p-values are not used to declare significance. This controls the family-wise error rate *within* each family at 0.05, not a single experiment-wide rate across all three families: treating the families as independent, the across-family error is approximately $1 - (1 - 0.05)^3 \approx 0.14$, so a comparison that is only marginally significant inside its family should not be read as protected at the 0.05 level once the three families are considered jointly. We keep per-family control deliberately, because the families answer distinct questions (degradation relative to the original, encoder-vs-encoder ranking, and pointwise ranking) and are reported and interpreted separately.

5.3.2. Pointwise Error Metrics

To evaluate the reconstruction methods, we use the following error metrics:

- **Mean Absolute Error (MAE):** This metric represents the average of the absolute differences between the original and reconstructed values. It provides a direct measure of the magnitude of the errors without considering their direction.
- **Mean Relative Error (MRE):** Calculated as

$$\text{MRE} = \frac{1}{n} \sum_{i=1}^{n} \frac{|x_i - \hat{x}_i|}{|x_i| + \varepsilon},$$

  where $\varepsilon = 10^{-8}$ prevents division by zero when accelerometer values are zero or close to zero. This metric normalizes the absolute error by the magnitude of the original signal.
- **Mean Squared Error (MSE):** By squaring each error before averaging, the MSE gives a higher penalty to larger errors, thereby emphasizing large deviations in the reconstruction.
- **Pearson Correlation Coefficient:** This statistical measure indicates the degree of linear correlation between the original and reconstructed signals. A coefficient close to 1 indicates that the reconstructed signal follows the original temporal pattern.

These metrics are computed for each accelerometer axis (Dim 0, Dim 1, and Dim 2) and then aggregated as a global mean, allowing axis-level errors and overall reconstruction quality to be reported separately for RP, MTF, GASF, and GADF.

5.3.3. Polynomial Trend Metrics

To evaluate the performance of the polynomial representation reconstruction, the following normalized metrics were computed on the cubic coefficient vectors. Each coefficient dimension was standardized using the mean and standard deviation estimated from the original windows before distances were calculated. For Mahalanobis distance, the covariance matrix was estimated from the standardized coefficient vectors of the original windows and regularized as $\Sigma_\lambda = \Sigma + 10^{-6} I$ before inversion.

- **Mean Euclidean Distance:** This metric calculates the average Euclidean distance between the standardized cubic coefficient vectors of the original and reconstructed signals. A lower value indicates a more accurate reconstruction.
- **Mean Manhattan Distance:** This metric represents the mean absolute difference (L1 norm) between the standardized cubic coefficient vectors, and it is sensitive to the overall magnitude of the differences.
- **Mean Cosine Similarity:** This metric evaluates the cosine of the angle between the standardized coefficient vectors, yielding a value close to 1 when the reconstructed polynomial follows the same direction as the original, regardless of magnitude differences.
- **Mean Mahalanobis Distance:** This metric measures the distance between the standardized coefficient vectors using Σ_λ^{-1}, taking into account the variance and correlation structure estimated from the original coefficient vectors. It provides a multidimensional assessment of the similarity between the original and reconstructed polynomials.

The quantitative results for these metrics are presented in Table 10 later in the document.

5.3.4. Classifier Evaluation

To test whether reconstructed windows retain task-relevant information, we evaluated them with several classifiers. These classifiers do not prove pointwise fidelity; instead, they indicate whether class-discriminative temporal structure remains available after reconstruction. The selection of classifiers was inspired by Ignatov et al. [19], who analyzed multiple classifiers on accelerometer-based activity recognition.

Specifically, we used a three-fold protocol with $n = 3$ partitions. Each fold holds out one partition for testing and uses the remaining two partitions for training, producing a 66.6%/33.3% train-test split per fold. The 8,172 instances correspond to triaxial windows, not to separate one-dimensional axes; therefore, each fold contains approximately 2,724 test windows. Reported metrics represent the average performance obtained across the three folds when comparing the original time series against their reconstructed counterparts.

A detailed overview of each classifier and its intended role is presented below:

- **Gated Recurrent Unit (GRU):** A recurrent neural network variant designed to effectively capture temporal dependencies in sequential data. Its gating mechanisms regulate the information flow, enabling recognition of complex patterns over time.
- **K-Nearest Neighbors (KNN):** An instance-based classifier that assigns labels based on the majority class of the closest k neighbors. It acts as a baseline to determine how well separated the reconstructed classes are within the feature space.
- **Convolutional Neural Network (CNN):** A deep neural network architecture optimized for extracting local temporal patterns and spatial relationships through convolutional filters. CNN effectively identifies discriminative motifs critical for activity classification.
- **Random Forest (RF):** An ensemble classifier comprising multiple decision trees that independently vote to determine the final class prediction. RF robustly captures nonlinear relationships and interactions among features.
- **Support Vector Machine (SVM):** This classifier finds separating hyperplanes in high-dimensional feature spaces and tests whether reconstructed windows remain separable under a margin-based model.
- **Long Short-Term Memory (LSTM):** A recurrent neural network designed to model long-term dependencies in sequential data. Its gating mechanisms test whether longer temporal correlations remain available after reconstruction.

The exact configuration of each classifier is summarised in Table 5. To make the neural-network results reproducible, the global random seed (Python, NumPy and TensorFlow) is fixed to 95; K-Nearest Neighbors and Support Vector Machine are deterministic given the training data, and Random Forest uses the same value as its `random_state`.

Table 5: Classifier configurations. Each model is trained once per fold on the original (un-reconstructed) windows and reused across all four encodings. Input is $(128, 3)$ for the neural models and the flattened 384-vector for the scikit-learn models; all neural models use the sparse categorical cross-entropy loss.

Classifier	Configuration	Seed
KNN	$k = 5$, uniform weights, Minkowski distance ($p = 2$)	—
SVM	RBF kernel, $C = 1.0$, $\gamma =$ scale	—
Random Forest	100 trees, Gini split, unlimited depth, `max_features`$= \sqrt{d}$	95
LSTM	LSTM(64) $\to$ Dropout(0.2) $\to$ LSTM(32) $\to$ Dropout(0.2) $\to$ Dense-softmax; Adam (10^{-3}), 20 epochs, batch 64	95
GRU	GRU(64) $\to$ Dropout(0.2) $\to$ GRU(32) $\to$ Dropout(0.2) $\to$ Dense-softmax; Adam (10^{-3}), 20 epochs, batch 64	95
CNN	Conv1D(196, kernel 16) + MaxPool(4) concatenated with per-channel statistical features (mean, variance, $\sum \lvert \cdot \rvert$, 10-bin histogram); Dense(1024) + Dropout(0.05) + Dense-softmax; $\ell_2 = 10^{-4}$; Adam (5×10^{-4}), 20 epochs, batch 64	95

6. Results

The results are presented from the most direct evidence to the most indirect: visual reconstruction, pointwise error tables, polynomial trend similarity, and downstream classification. This order makes clear when a method reconstructs the signal itself and when it merely preserves information useful for a classifier.

6.1. Qualitative Reconstruction Analysis

6.1.1. Artificial Time Series

For our first experiment, we construct a synthetic time series $\mathbf{w} \in \mathbb{R}^{N\times 3}$ of length $N = 128$ by sampling an underlying parameter t uniformly over $[0, \pi]$. Concretely, define

$$t_i = \frac{i}{N-1}\,\pi, \qquad i = 0, 1, \ldots, N-1,$$

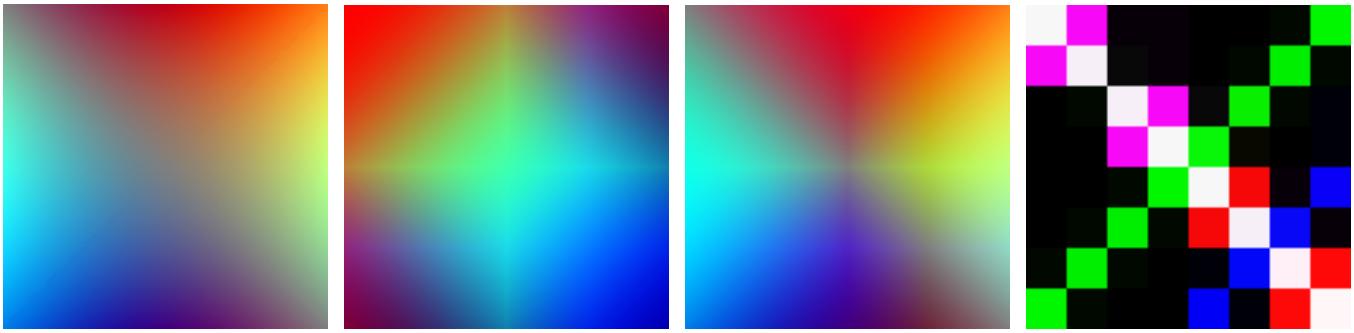
and let each vector $\mathbf{w}_i = [w_{i,0},\, w_{i,1},\, w_{i,2}]^\top$ be given by

$$w_{i,0} = \cos(t_i), \qquad w_{i,1} = \sin(t_i), \qquad w_{i,2} = \sin(t_i) \;-\; \cos(t_i).$$

Thus the full data matrix is

$$\mathbf{w} = \begin{bmatrix} \cos(t_0) & \sin(t_0) & \sin(t_0) - \cos(t_0) \\ \cos(t_1) & \sin(t_1) & \sin(t_1) - \cos(t_1) \\ \vdots & \vdots & \vdots \\ \cos(t_{N-1}) & \sin(t_{N-1}) & \sin(t_{N-1}) - \cos(t_{N-1}) \end{bmatrix}.$$

This three-dimensional signal, composed of a pure cosine channel, a pure sine channel, and their difference, serves as a controlled sanity check for the RP, GASF, GADF, and MTF reconstructions. It is not intended to represent the full WISDM distribution; instead, it provides a visually interpretable case where phase, amplitude, and orientation errors are easy to detect. To keep the evidence separated, every figure in this subsection refers only to the artificial control signal. Figure 9 shows the image encodings of the three artificial channels under each of the four techniques (RP is 127×127 because of the $N - 1$ pair embedding; the others are 128×128).



(a) RP (b) GASF (c) GADF (d) MTF

Figure 9: Image encodings of the same artificial three-channel signal (cosine, sine, sine−cosine) under each technique.

After generating the encoded images, we decode them and compare the reconstructed series graphically.

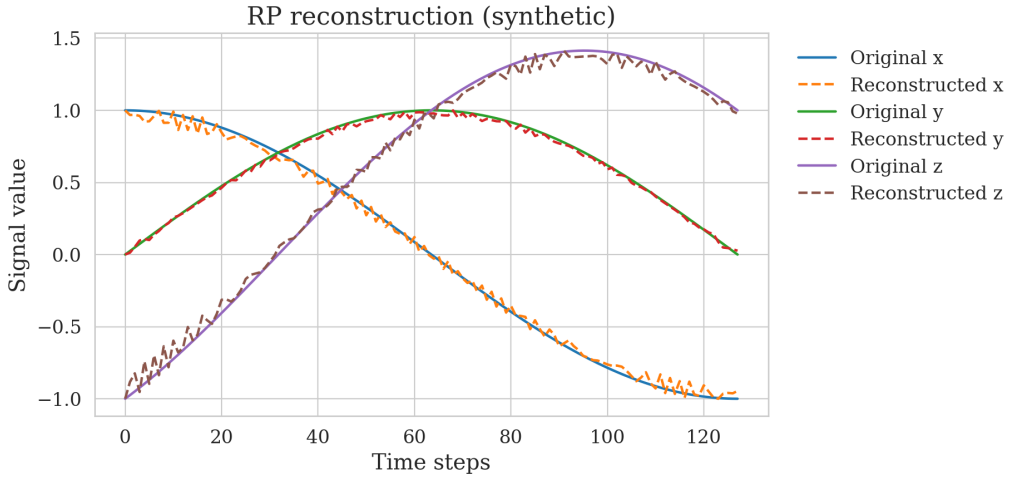


Figure 10: RP reconstruction on the artificial triaxial time series.

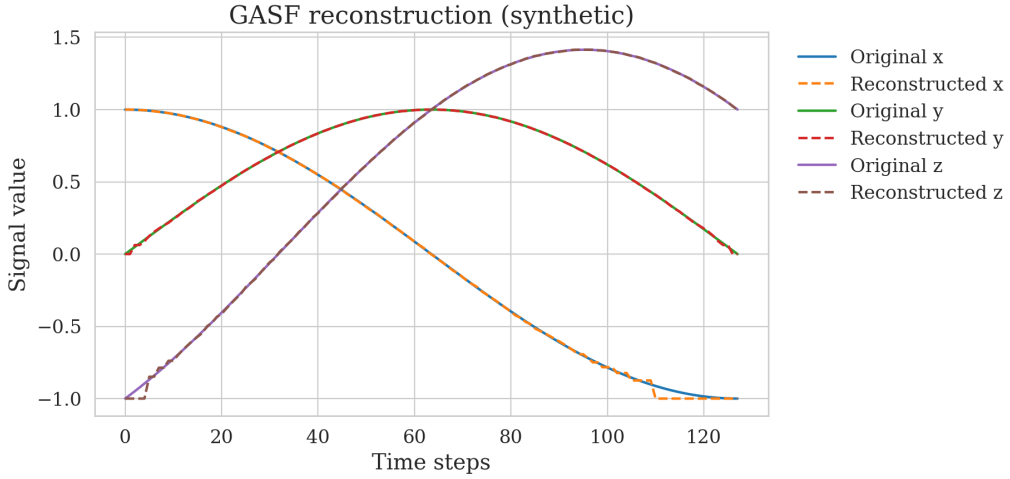


Figure 11: GASF reconstruction on the artificial triaxial time series.

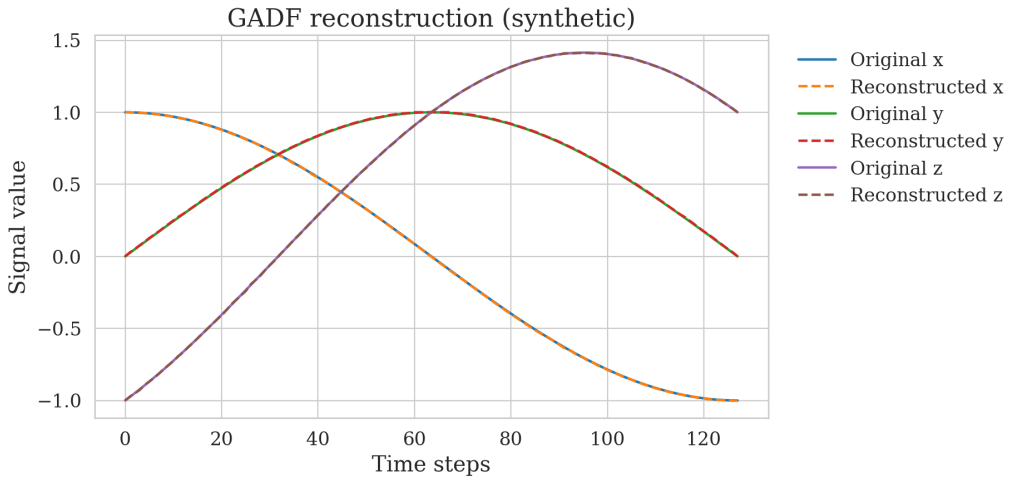


Figure 12: GADF reconstruction on the artificial triaxial time series.

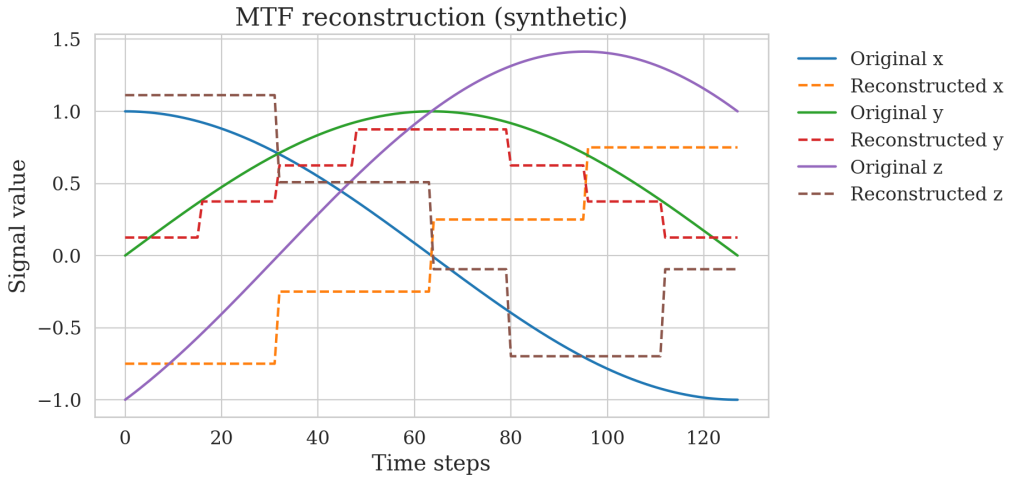
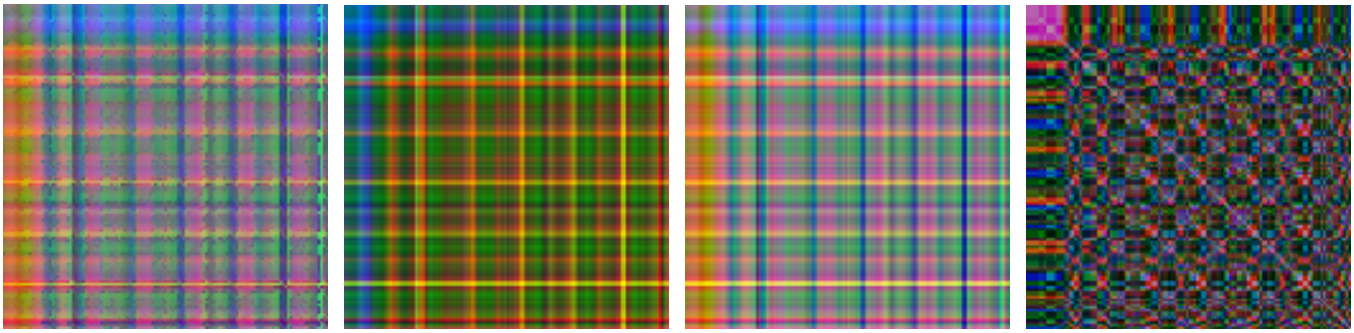


Figure 13: MTF reconstruction on the artificial triaxial time series.

After encoding the artificial time series (Figure 9), we reconstruct each representation and overlay it with the original signal. In this controlled example, the RP comparison (Figure 10) shows close shape recovery with small amplitude deviations. The GASF comparison (Figure 11) is visually closest to the original in all three channels under the tested normalization. The GADF comparison (Figure 12) is likewise near-perfect: the multi-pivot, boundary-anchored reconstruction overlays the cosine, sine and sin−cos channels almost exactly, confirming that the restricted-branch inverse recovers the synthetic signal up to image quantisation. By contrast, the MTF comparison (Figure 13) reveals two limitations that also matter in real data: possible orientation inversion and discretization errors caused by the four transition bins.

6.1.2. WISDM Sample `1600×0`

To assess encoding–decoding fidelity on real sensor data, we selected sample `1600×0` from fold 0 of the WISDM training set (class 0, Walking, $N = 128$ samples). This example is used for qualitative interpretation only; aggregate claims are based on the quantitative tables that follow. Figure 14 shows the encoded images of *the same time-series window* under each of the four encoding techniques, with the three accelerometer axes (x, y, z) mapped to the red/green/blue channels of the image. The RP image is 127×127 because the modified definition $s_j = (x_j, x_{j+1})$ embeds the window into $N - 1$ adjacent states; GASF, GADF, and MTF are 128×128.



(a) Recurrence Plot (RP) (b) GASF (c) GADF (d) Markov Transition Field (MTF)

Figure 14: The four image encodings of the same WISDM window `1600×0` (fold-0 train, Walking). Per-axis (x,y,z) intensities are stacked as the (R,G,B) channels of each image.

After encoding, each image is decoded back into a three-axis time series and overlaid on the original signal. Figures 15–18 present the global comparison across all four encoders (RP, GASF, GADF, MTF).

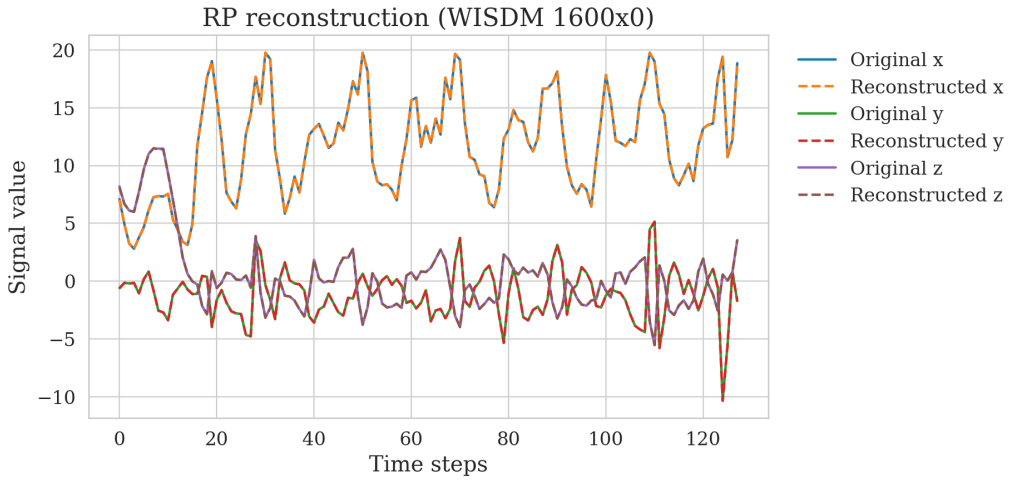


Figure 15: Original vs. RP reconstruction for sample `1600×0`.

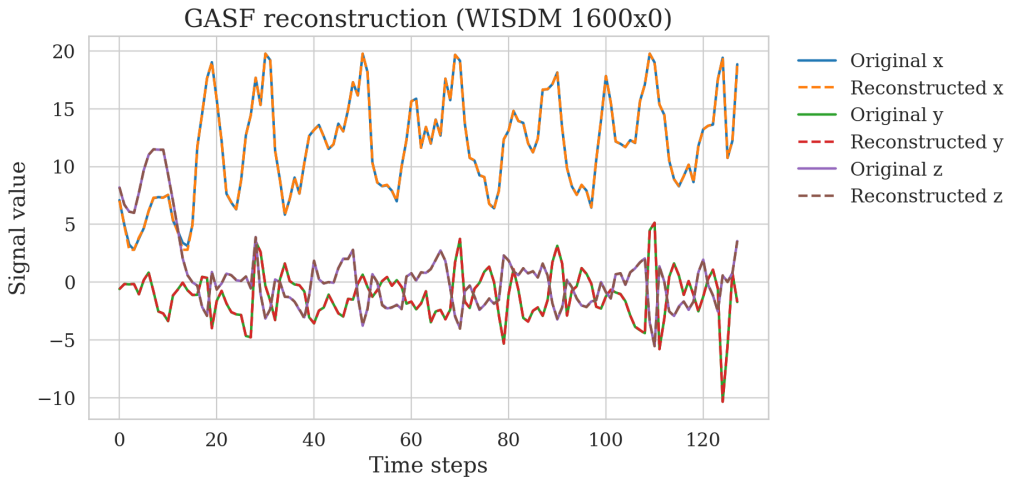


Figure 16: Original vs. GASF reconstruction for sample `1600×0`.

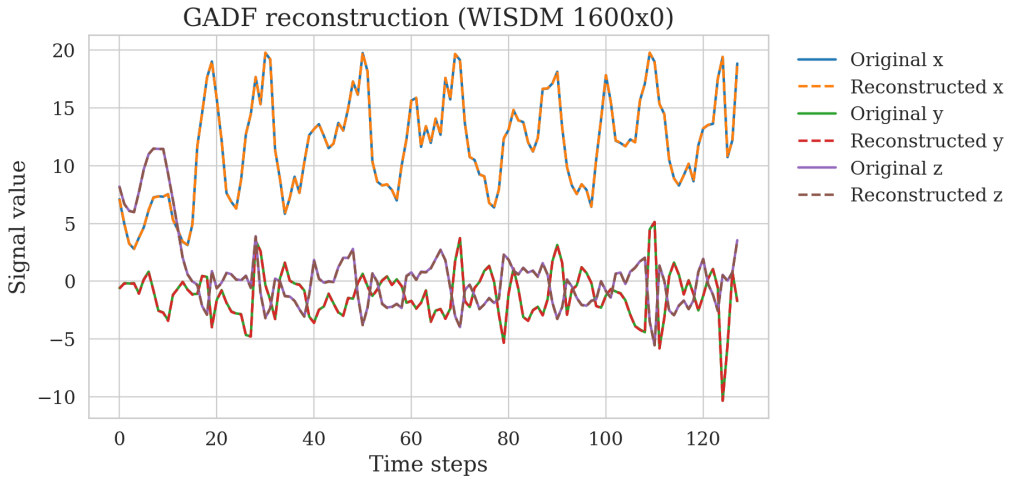


Figure 17: Original vs. GADF reconstruction for sample `1600×0`.

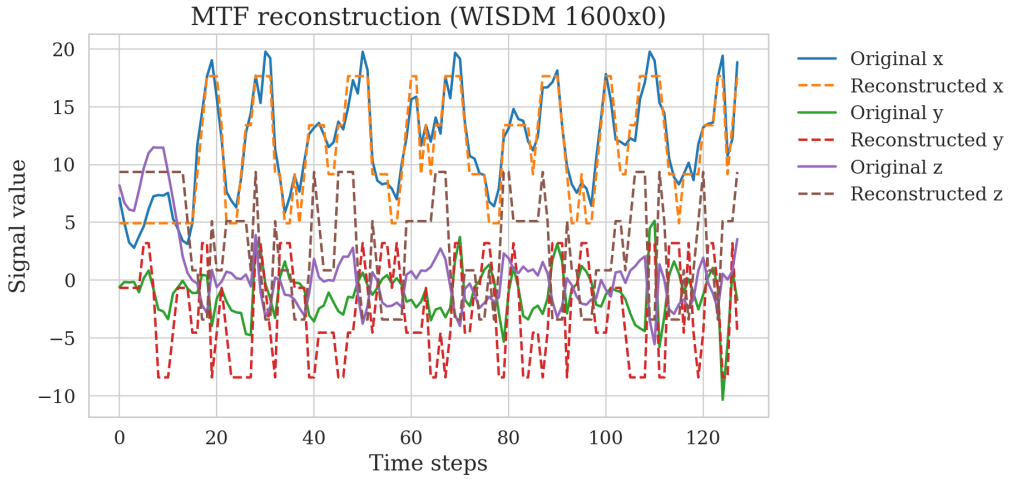


Figure 18: Original vs. MTF reconstruction for sample `1600×0`.

Across the four evaluated methods, GASF (Fig. 16) produces the closest visual reconstruction under the tested protocol, with decoded signals nearly indistinguishable from the original. GADF (Fig. 17) is visually on par with GASF: its multi-pivot, boundary-anchored decoder recovers both the global shape and the short-range variation, consistent with its near-original pointwise error and the absence of any detected classifier degradation in Section 7.5. RP (Fig. 15) recovers part of the temporal morphology but remains less stable than the GAF family because its graph reconstruction still has reflection and length-restoration ambiguity. MTF (Fig. 18), despite using only four quantised transition levels, captures the general shape but shows larger deviations in segments where neighboring transition states are difficult to order.

For clarity, Figure 19–21 split the MTF comparison per axis:

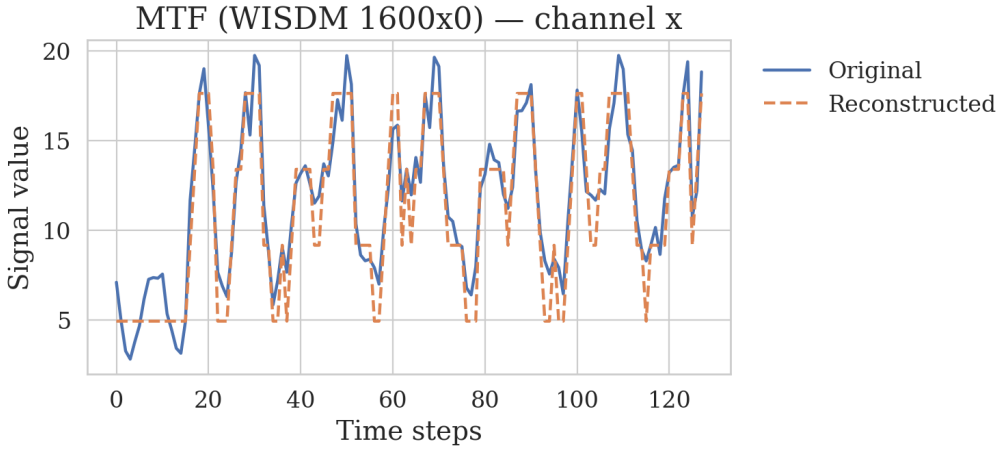


Figure 19: Original vs. MTF reconstruction for sample `1600×0`, axis x.

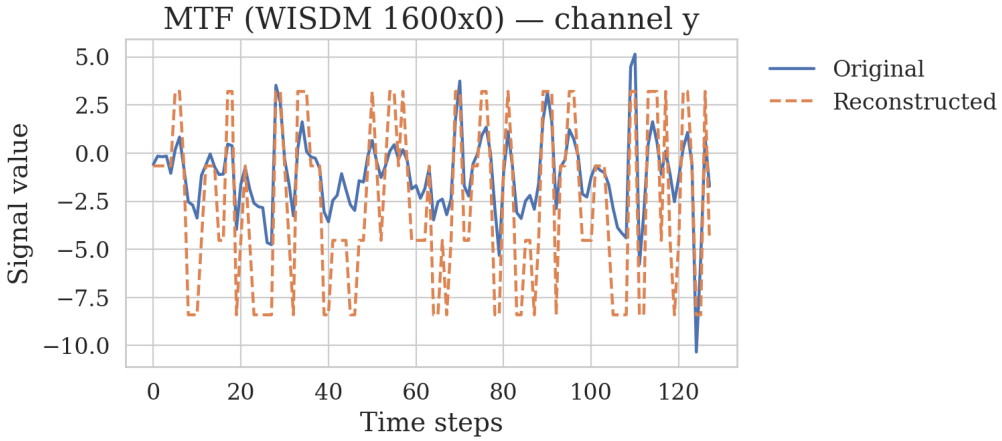


Figure 20: Original vs. MTF reconstruction for sample `1600×0`, axis y.

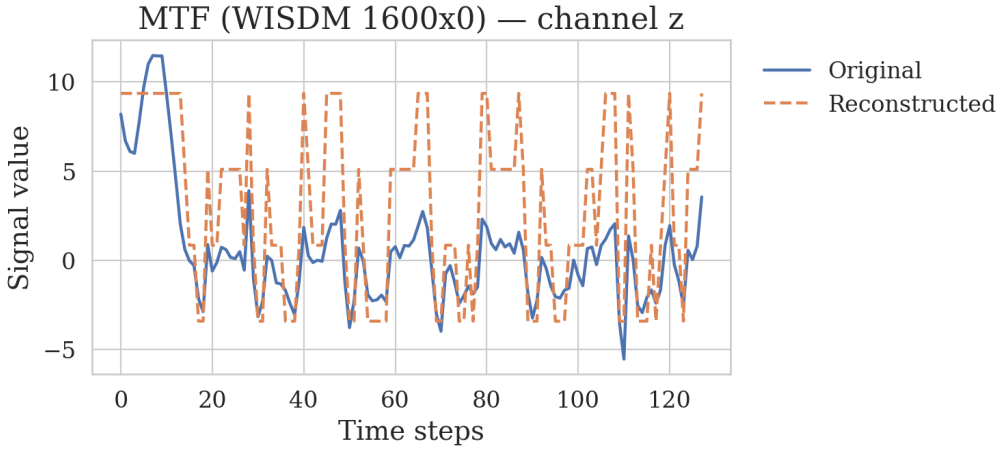


Figure 21: Original vs. MTF reconstruction for sample `1600×0`, axis z.

6.2. Quantitative Reconstruction Results

6.2.1. Pointwise Error Metrics

Unless otherwise specified, all error calculations were performed on the raw (unnormalized) time series of the window-level fold-0 test split, in the dataset's native accelerometer units (m/s^2), so as to quantify the absolute magnitude of reconstruction errors in the context of the data's native amplitude range.

Table 6: Recurrence Plot Errors

Metric	Dim 0	Dim 1	Dim 2	Global Mean
Mean Absolute Error	0.0147 ± 0.0198	0.0181 ± 0.0251	0.0114 ± 0.0144	0.0148 ± 0.0204
Mean Relative Error	0.0036 ± 0.0056	0.0051 ± 0.0061	0.0054 ± 0.0066	0.0047 ± 0.0062
Mean Squared Error	0.0009 ± 0.0048	0.0014 ± 0.0048	0.0005 ± 0.0018	0.0009 ± 0.0041
Pearson Correlation Coefficient	0.99998 ± 0.00008	0.99998 ± 0.00004	0.99998 ± 0.00007	0.99998 ± 0.00007

The proposed decoder recovers the two-dimensional state geometry directly from the recovered distances and does not use the recurrence-rate parameter ρ of the earlier graph-based variant, so no recurrence-rate sensitivity analysis applies.

Table 7: Markov Transition Fields Errors (Fiedler-vector decoder, $Q = 4$, window-level fold 0; the 3-fold-pooled equivalent is the $Q = 4$ row of Table 16, MAE 3.95 and Pearson 0.013).

Metric	Dim 0	Dim 1	Dim 2	Global Mean
Mean Absolute Error	4.24 ± 5.20	4.15 ± 5.36	3.56 ± 4.66	3.98 ± 5.09
Mean Relative Error	0.92 ± 1.26	1.26 ± 1.61	1.48 ± 1.51	1.22 ± 1.49
Mean Squared Error	59.78 ± 124.70	62.16 ± 162.86	47.33 ± 152.81	56.42 ± 147.82
Pearson Correlation Coefficient	-0.03214 ± 0.84308	0.04466 ± 0.83559	-0.00386 ± 0.82390	0.00289 ± 0.83483

Table 8: Gramian Angular Summation Field Errors

Metric	Dim 0	Dim 1	Dim 2	Global Mean
Mean Absolute Error	0.0233 ± 0.0236	0.0193 ± 0.0220	0.0181 ± 0.0215	0.0202 ± 0.0225
Mean Relative Error	0.0055 ± 0.0066	0.0045 ± 0.0045	0.0082 ± 0.0105	0.0060 ± 0.0078
Mean Squared Error	0.0049 ± 0.0100	0.0049 ± 0.0130	0.0030 ± 0.0108	0.0042 ± 0.0114
Pearson Correlation Coefficient	0.99987 ± 0.00017	0.99990 ± 0.00029	0.99987 ± 0.00033	0.99988 ± 0.00027

Table 9: Gramian Angular Difference Field Errors

Metric	Dim 0	Dim 1	Dim 2	Global Mean
Mean Absolute Error	0.0157 ± 0.0184	0.0168 ± 0.0208	0.0144 ± 0.0194	0.0156 ± 0.0196
Mean Relative Error	0.0038 ± 0.0055	0.0054 ± 0.0067	0.0071 ± 0.0088	0.0054 ± 0.0073
Mean Squared Error	0.0007 ± 0.0019	0.0009 ± 0.0023	0.0007 ± 0.0026	0.0008 ± 0.0023
Pearson Correlation Coefficient	0.99999 ± 0.00002	0.99999 ± 0.00002	0.99999 ± 0.00002	0.99999 ± 0.00002

6.2.2. Polynomial Trend Comparison

Table 10: Normalized Coefficient Metrics

Metric	RP	GASF	GADF	MTF
Mean Euclidean Distance	0.15952	0.00079	0.00070	0.22991
Mean Manhattan Distance	0.16578	0.00084	0.00072	0.24219
Mean Cosine Similarity	0.92819	0.99994	1.00000	0.90817
Mean Mahalanobis Distance	1.89103	0.01042	0.00728	2.88047

6.3. Downstream Classification Results

The full per-class precision, recall, F1, and support for all six classifiers — on the original validation windows and on each reconstructed signal (RP, MTF, GASF, GADF) — are reported in Appendix A; the body retains only the compact accuracy comparison (Table 18) needed to interpret the differences among reconstruction methods. Those numbers are averaged across all 3 folds of the stratified window-level protocol (mean ± std, ∼2 724 test windows per

fold) and are scenario-B results in Table 4: all six classifiers share the same trained model across encodings, fitted once on the original training set, so only the prediction input changes across the GASF/GADF/RP/MTF columns.

6.4. Grouped Subject-Out Robustness

The stratified window-level 3-fold protocol used so far is a class-stratified split at the window level: because the grouping key is effectively unique per window, windows from the same subject – and indeed from the same recording – can fall on both sides of a fold. This is a common configuration for within-distribution evaluation and the right test for whether a reconstruction preserves task-relevant temporal structure within the observed population. It is not, however, a test of whether the model generalises to *unseen subjects* – the regime that matters most for wearable deployment. We therefore replicate the full evaluation under a grouped subject-out 3-fold protocol: the 51 WISDM subjects are partitioned by StratifiedGroupKFold into three disjoint groups of 16, 18 and 17 test subjects respectively, and the test set of one fold never shares a subject with either of its training folds. We use the term subject-out rather than strict leave-one-subject-out, since each fold holds out a disjoint group of subjects rather than a single subject. Encoding, reconstruction, and the six-classifier panel are exactly the same as for the window-level protocol; only the train/test indices change. Each subject-out fold contains ∼2 724 test windows so the pooled paired-sample size for inferential tests is again $N = 8\,172$.

Table 11: Headline accuracies under the grouped subject-out 3-fold protocol (mean ± std across folds 0, 1, 2; ∼2 724 test windows per fold drawn from 16–18 held-out subjects). Δacc is the accuracy of the reconstructed signal minus the accuracy of the original signal under the same classifier, averaged over the same three folds.

Classifier	**Orig.**	**GASF**	**GADF**	**RP**	**MTF**
GRU	0.811 ± 0.026	0.811 ± 0.026 (+0.000)	0.811 ± 0.026 (+0.000)	0.811 ± 0.026 (+0.000)	0.702 ± 0.025 (−0.109)
KNN	0.504 ± 0.035	0.504 ± 0.034 (−0.000)	0.505 ± 0.034 (+0.000)	0.504 ± 0.035 (−0.000)	0.465 ± 0.039 (−0.040)
CNN	0.796 ± 0.036	0.796 ± 0.036 (−0.000)	0.796 ± 0.036 (−0.000)	0.794 ± 0.049 (−0.003)	0.686 ± 0.035 (−0.110)
Random Forest	0.786 ± 0.039	0.787 ± 0.039 (+0.001)	0.787 ± 0.039 (+0.000)	0.786 ± 0.039 (−0.001)	0.689 ± 0.026 (−0.098)
Support Vector Machine	0.738 ± 0.020	0.738 ± 0.020 (+0.000)	0.738 ± 0.020 (−0.001)	0.739 ± 0.020 (+0.000)	0.665 ± 0.026 (−0.073)
LSTM	0.744 ± 0.029	0.744 ± 0.029 (+0.000)	0.744 ± 0.028 (−0.000)	0.744 ± 0.028 (−0.000)	0.672 ± 0.035 (−0.072)

Table 12: Per-fold accuracies on the grouped subject-out 3-fold protocol. Folds split the 51 WISDM subjects into disjoint groups of 16, 18 and 17 test subjects respectively; the test set of one fold never shares a subject with either of its training folds.

	Orig.			**GASF**			**GADF**			**RP**			**MTF**		
Classifier	f0	f1	f2	f0	f1	f2	f0	f1	f2	f0	f1	f2	f0	f1	f2
GRU	0.848	0.791	0.794	0.848	0.791	0.794	0.847	0.791	0.794	0.848	0.791	0.794	0.732	0.671	0.702
KNN	0.551	0.495	0.467	0.550	0.494	0.468	0.550	0.496	0.467	0.551	0.495	0.466	0.515	0.458	0.421
CNN	0.847	0.779	0.764	0.847	0.778	0.764	0.847	0.778	0.764	0.859	0.740	0.782	0.735	0.670	0.654
Random Forest	0.842	0.753	0.764	0.841	0.754	0.766	0.842	0.754	0.765	0.840	0.751	0.765	0.726	0.666	0.674
Support Vector Machine	0.757	0.711	0.748	0.757	0.710	0.748	0.756	0.709	0.747	0.757	0.711	0.748	0.701	0.639	0.654
LSTM	0.779	0.708	0.745	0.779	0.708	0.745	0.778	0.709	0.745	0.777	0.708	0.745	0.719	0.633	0.664

Three observations follow from Tables 11 and 12, read together with the window-level-vs-subject-out summary that opens the discussion (Table 18):

1. **The encoder ranking is preserved under subject-out evaluation, including the RP/MTF lens inversion of Section 7.5.** This is the load-bearing robustness check for the structural argument: the window-level orderings are not artefacts of allowing the model to memorise per-subject calibration. Under grouped subject-out validation the absolute accuracies drop by 0.06–0.14 as expected (the model has never seen the test subjects), and GASF and GADF remain statistically tied and near-original on every individual classifier of every fold ($p_{\text{Holm}} = 1.0$ for both Original-vs-GASF and Original-vs-GADF). The RP/MTF inversion, however, is protocol-sensitive: once the tuned $\rho = 0.60$ lifts RP, MTF significantly outranks RP under subject-out only on Random Forest and LSTM ($p_{\text{Holm}} = 6.6\times10^{-19}$ and 2.8×10^{-9}), while GRU, CNN and SVM become statistical ties and KNN reverses in RP's favour ($p_{\text{Holm}} = 2.7\times10^{-70}$). The strong window-level inversion therefore softens under the harder subject-out split, which is itself informative: the RP/MTF gap is the least stable feature of the ranking, whereas the near-original GASF/GADF top tier is not a within-distribution artefact.
2. **GASF shows no detected degradation from the original under this paired test.** The paired McNemar test on the pooled subject-out sample fails to reject Original-vs-GASF in all six classifiers ($p_{\text{Holm}} = 1.00$ for all six); the absolute subject-out accuracy drop of GASF relative to the original is bounded by $|\Delta\text{acc}| \leq 0.001$ across the six classifiers, identical to the window-level behaviour. This replicates the near-equivalence pattern of Section 7.5, while still stopping short of a formal equivalence or information-preservation claim.

3. **Under subject-out validation, RP is the only reconstruction that strictly improves on the original, and only for KNN.** On KNN, RP reaches 0.549 ± 0.029 while the original signal only reaches 0.504 ± 0.035 (+0.044 acc; paired McNemar $p_{\text{Holm}} = 5.2 \times 10^{-28}$ in RP's favour); GASF and GADF, being near-original reconstructions, simply track the raw signal (0.504 and 0.505, $p_{\text{Holm}} = 1.0$ against the original). The RP gain is driven by the tuned recurrence rate $\rho = 0.60$ and also holds under the window-level split (0.686 vs 0.642). This is consistent with a regularization or low-pass hypothesis: the same smoothing that costs RP accuracy on the high-ceiling classifiers becomes beneficial for the instance-based KNN classifier under the harder subject-out split. However, a dedicated frequency-domain analysis – e.g., original vs. reconstructed power spectral density, high-frequency energy ratios, and smoothed-original baselines such as moving average, Savitzky–Golay or Butterworth filters – is required before interpreting RP as a denoising mechanism.

6.5. GADF Decoder Reliability: Exactness, Anchor, and Pivot Ensemble

We verify the two numeric choices of the GADF decoder — the amplitude restriction α and the number of averaged central pivots n_p — and its deterministic boundary-row rule, at the operating point $\alpha = 0.7$ on 1 500 WISDM windows (4 500 axis-series). The floating-point inverse is numerically *exact* there (MAE $< 10^{-6}$): the closed-form reconstruction carries no algorithmic error, so the entire GADF error reported elsewhere in this paper is 8-bit PNG quantisation (physical MAE 0.0158, Pearson 1.0000), not decoder failure. The boundary anchor never mis-identifies (0% failure across the 4 500 axis-series), because per-window min–max normalisation forces both amplitude extremes to be present, so the extreme-mean row is always a genuine boundary. We fix $\alpha = 0.7$ as the largest value below the branch-uniqueness threshold $1/\sqrt{2} \approx 0.7071$ (Section 4.4), which maximises the effective use of the 8-bit intensity range while keeping every pairwise angular difference inside the principal branch of arcsin.

Pivot-ensemble size. Averaging the reconstruction over the n_p most-central rows, rather than trusting a single pivot, is the second numeric choice. Table 13 sweeps n_p on the same PNG round trip at $\alpha = 0.7$: the pointwise error falls monotonically — more than halving from a single pivot (0.0353) to five (0.0158, the value used throughout the paper) and improving further to 0.0119 at $n_p = 9$ — at essentially constant cost (1.54–1.59 ms per axis-series), since each extra pivot is one more $O(N)$ pass over an already-loaded matrix. We fix $n_p = 5$ as a conservative default that captures most of the averaging gain; a larger ensemble would lower the reported GADF error further at negligible extra cost, but n_p is kept fixed so the headline numbers are not tuned on the evaluation windows.

Table 13: GADF pivot-ensemble sweep on the same 1 500 WISDM windows (4 500 axis-series) at $\alpha = 0.7$. n_p is the number of averaged central pivot rows; MAE is the 8-bit PNG round-trip error in physical units; time is per axis-series. The default used everywhere else in the paper is $n_p = 5$.

n_p	1	3	5	7	9
MAE (8-bit PNG)	0.0353	0.0206	**0.0158**	0.0134	0.0119
Time (ms/axis)	1.54	1.55	1.57	1.58	1.59

6.6. RP Decoder Ablation: Inverse Normalisation and Signed Orientation Selection

Two ingredients make the RP decoder near-invertible, and Table 14 isolates each on 100 stratified window-level fold-0 test windows. The first is the *inverse normalisation*: decoding the stored image with the RP matrix's own range rather than the signal amplitude range (Step 1). The second is the *signed orientation selection* (Step 5), which fixes the residual reflection of the two-dimensional embedding. We also report EVR_2, the fraction of the multidimensional-scaling spectrum captured by the two leading positive dimensions — a direct diagnostic of whether the recovered distances still form a two-dimensional Euclidean geometry.

Table 14: RP decoder ablation on 100 stratified window-level fold-0 test windows; MAE (raw) and Pearson are averaged over the three axes, and EVR_2 is the fraction of the MDS spectrum in the two leading positive dimensions. All rows use the RP-range inverse normalisation except where noted.

Variant	MAE	Pearson	EVR_2
DIRECT-2D-S (proposed decoder)	**0.010**	**1.000**	0.947
with signal-range decode (uncorrected)	1.047	0.959	0.218
without signed orientation selection	2.033	0.794	0.947
Prior one-dimensional graph decoder (Jaccard)	2.177	0.918	0.730

Both ingredients are decisive. Decoding with the signal amplitude range instead of the RP range raises the MAE from 0.010 to 1.047 and, more tellingly, collapses EVR_2 from 0.95 to 0.22: the affine scale mismatch turns the recovered magnitudes into a matrix that is no longer two-dimensionally Euclidean, so multidimensional scaling can

no longer place the states. Removing the signed orientation selection leaves the reflection unresolved and raises the MAE to 2.033. The prior one-dimensional graph decoder — which projected each state onto a single MDS axis and re-augmented the length with a per-window mean — reaches only MAE = 2.18 even with the corrected normalisation, confirming that recovering the full two-dimensional state geometry, not just a scalar projection, is what makes the signal recoverable.

6.7. MTF Decoder Ablation: Ordering and Metadata

The MTF decoder has two design choices we need to defend empirically: the bin-ordering algorithm (cosine-pivot heuristic vs Fiedler-vector spectral) and the bin count Q. We ablate both on the stratified window-level 3-fold protocol against the same six classifiers used in the rest of the paper.

Decoder ablation (at fixed $Q = 4$). We compare four decoder configurations on the same encoded MTF images (Table 15): the cosine-pivot ordering heuristic, Fiedler-vector ordering, Fiedler-vector ordering with stored bin-midpoint metadata, and Fiedler-vector ordering with both midpoint metadata and a sign-bit. The encoder is held constant. Bin midpoints (storing Q floats per axis, the median of each quantile bin) and a sign-bit (one bit per axis to disambiguate the mirror symmetry) are *not* part of the side-information contract used for the headline numbers; they characterize the achievable floor that any MTF decoder operating on top of the standard MTF image cannot beat.

Table 15: MTF decoder ablation at $Q = 4$, 100 stratified window-level fold-0 test windows ($T = 128$); pointwise MAE and Pearson averaged across the three accelerometer axes.

Decoder variant	MAE	Pearson
Cosine-pivot heuristic	6.493	+0.023
Fiedler ordering	5.060	−0.008
Fiedler + midpoint metadata	3.926	+0.013
Fiedler + midpoint + sign-bit	2.603	+0.465

The Fiedler ordering alone, with no extra metadata, drops the cosine-pivot pointwise MAE by 22% (6.49 → 5.06). Adding Q bin-median floats per axis (comparable in cost to the optional per-axis $(\min, \max)$ calibration used only for physical-scale recovery) drops it to 3.93; adding a single sign-bit per axis – which resolves the global reversal ambiguity of the inferred Fiedler ordering, though not the many-to-one quantisation and transition-profile ambiguities intrinsic to MTF – drops it further to 2.60. We use *Fiedler ordering without midpoint or sign-bit metadata* for every MTF number elsewhere in the paper because it keeps the side-information contract identical to the cosine-pivot heuristic, and characterizes what an MTF decoder can achieve without extra metadata beyond the standard optional per-axis physical-scale calibration.

Bin-count ablation across Q. The bin count Q is the parameter most often discussed in the MTF literature. We sweep $Q \in \{2, 4, 6, 8, 10, 16\}$ across the full stratified window-level 3-fold test set ($N = 8\,172$ pooled windows; approximately 2 724 per fold) and re-validate the six-classifier panel on each reconstruction, using the Fiedler decoder uniformly. Table 16 gives the result; the rows are sorted by Q to make the location of the minimum explicit.

Table 16: MTF bin-count ablation on the stratified window-level 3-fold protocol ($N = 8\,172$ pooled test windows). Pointwise MAE and Pearson are averaged across the three accelerometer axes; classifier accuracies are 3-fold means. The minimum MAE, maximum Pearson, and per-classifier optimum are highlighted in bold. The Fiedler-vector decoder is used uniformly; only Q varies.

Q	**MAE**	**Pearson**	KNN	SVM	RF	LSTM	GRU	CNN
2	4.11	+0.002	0.490	**0.705**	**0.762**	0.756	**0.771**	**0.774**
4	**3.95**	**+0.013**	**0.559**	0.697	0.757	**0.757**	0.765	0.773
6	4.01	+0.004	0.556	0.697	0.757	0.755	0.761	0.770
8	4.03	−0.001	0.539	0.695	0.753	0.757	0.764	0.765
10	4.07	+0.000	0.527	0.693	0.754	0.756	0.762	0.758
16	4.13	−0.014	0.520	0.697	0.755	0.757	0.762	0.753

Three observations matter:

1. **Pointwise fidelity is jointly maximised at $Q = 4$.** Both MAE and Pearson reach their optimum at $Q = 4$ (3.95, +0.013); every other choice of Q strictly degrades both metrics. The amplitude of the variation is modest (MAE moves in $[3.95, 4.13]$ across the swept range) because the Fiedler decoder is dominated by the intra-bin variance regardless of Q, but the location of the minimum is unambiguous.

2. **The amplitude-sensitive KNN classifier peaks at $Q = 4$.** KNN – the only classifier in our panel that measures local geometric similarity in the flattened sample space – improves by +6.9 percentage points going from $Q = 2$ to $Q = 4$ ($0.490 \rightarrow 0.559$) and then monotonically degrades down to 0.520 at $Q = 16$. CNN, which extracts local temporal patterns through learned convolutional filters, is similar at $Q = 2$ and $Q = 4$ (0.774 vs 0.773) and decreases moderately for larger Q (to 0.753 at $Q = 16$). KNN, the classifier that most exploits fine-grained amplitude structure, is thus the one that clearly pays for under- or over-quantisation.

3. **Coarse-pattern classifiers are nearly insensitive to Q.** SVM, Random Forest, LSTM and GRU vary by at most 0.012 accuracy across the entire $Q \in \{2, \ldots, 16\}$ sweep, with their (statistically negligible) optimum located at $Q = 2$. These four classifiers absorb most of the discriminative signal through bin-transition counts rather than through quantised amplitude shape, which is why they tolerate the coarsest quantisation. Their indifference to Q also explains why MTF as a representation is robust to the choice of Q: in the regime where most of the activity signal lives in the transition structure (high-ceiling neural classifiers), MTF's pre-quantisation step is not the bottleneck.

Taken together, the two ablations defend the $Q = 4$ default empirically: it is the unique value that minimises pointwise MAE, maximises pointwise Pearson, and maximises the amplitude-sensitive KNN accuracy (CNN being essentially flat, within 0.001 of its $Q = 2$ value), while costing at most 0.012 accuracy on the four classifiers that are essentially flat across Q.

7. Discussion

The four encodings have now been measured under the same WISDM protocol – $N = 128$ windows, full stratified window-level 3-fold evaluation ($N = 8\,172$ pooled test windows; $\sim$2 724 per fold), six classifiers fit *once per fold* on the originals and evaluated on every reconstruction – so the comparison no longer mixes one fully evaluated method against partially evaluated ones and is no longer restricted to a single split. We organise the discussion around four lenses (visual reconstruction, pointwise error, polynomial-trend similarity, and downstream classification) and ask, in turn, where the four encodings sit on each lens and whether the rankings agree. Table 17 states the role we assign to each operator after this round of evaluation; the compact classifier summary that underpins it, side by side for the window-level and subject-out protocols, is Table 18, the single most useful read of the comparison.

Table 17: Interpretation of each encoding after the unified $N = 128$ evaluation.

Method	Strength	Main Limitation	Empirical role
RP	Recurrence geometry and temporal morphology	Threshold, sign-flip ambiguity after classical MDS, and $N-1$ adjacent-state shrinkage	Pointwise third, classifier fourth: partially amplitude-preserving but classifier-fragile
GASF	Pointwise reconstruction through diagonal information (closed form)	Branch ambiguity outside $[0, \pi/2]$; min–max needed only for physical units	Near-original on every lens; tied with GADF as the two near-original decoders
GADF	Directional angular differences with $O(N^2)$ boundary-anchored angular inverse ($O(N)$ reconstruction after $O(N^2)$ row statistics)	Branch, sign convention, and angular anchor; min–max only for physical units	Near-original decoder, statistically tied with GASF in classification and marginally tighter on pointwise error; closed-form once the anchor row is fixed
MTF	Coarse transition dynamics, partly robust to quantisation; spectral Fiedler-vector ordering halves the cosine-pivot pointwise MAE	Intra-bin magnitude loss and global mirror ambiguity (a single sign bit would resolve it but is not stored here)	Pointwise fourth, classifier third on 4/6 (SVM tie, KNN to RP): classifier-friendly representation that exposes the lens inversion of Section 7.6

Table 18: Stratified window-level vs grouped subject-out accuracies (mean across 3 folds) — the compact summary that the rest of this section unpacks. The grouped subject-out protocol holds out 16–18 subjects per fold with zero overlap, so it is the stricter subject-generalisation test. The top of the ranking, the near-original GASF/GADF pair, is preserved under both sampling schemes. The bottom two encodings invert by evaluation lens: on the classifier accuracies shown here MTF significantly outranks RP on four of the six window-level classifiers (SVM a tie, KNN reversing to RP) and softens further under subject-out (significant only on Random Forest and LSTM), whereas on the pointwise reconstruction lens the order is RP $\succ$ MTF — the rank inversion discussed in the text. Absolute accuracies are lower under grouped subject-out validation because the model has not seen any window from the test subjects.

	Orig.		GASF		GADF		RP		MTF	
Classifier	Window	Subject-out	Window	Subject-out	Window	Subject-out	Window	Subject-out	Window	Subject-out
GRU	0.923	0.811	0.923	0.811	0.923	0.811	0.749	0.702	0.765	0.702
KNN	0.642	0.504	0.642	0.504	0.643	0.505	0.686	0.549	0.559	0.465
CNN	0.907	0.796	0.907	0.796	0.907	0.796	0.747	0.691	0.773	0.686
Random Forest	0.891	0.786	0.891	0.787	0.892	0.787	0.706	0.648	0.757	0.689
Support Vector Machine	0.801	0.738	0.801	0.738	0.801	0.738	0.694	0.657	0.697	0.665
LSTM	0.869	0.744	0.868	0.744	0.869	0.744	0.714	0.641	0.757	0.672

Complete per-class precision, recall, and F1-score results are provided in Appendix A.

7.1. Visual Reconstruction Trade-offs

Figures 15–18 show the four reconstructions overlaid on the original triaxial window. GASF (Fig. 16) is visually indistinguishable from the original on all three axes, including high-frequency wiggles, confirming that the diagonal half-angle rule recovers the normalised shape and that the optional physical-scale calibration restores the native amplitude. GADF (Fig. 17) is likewise visually indistinguishable from the original, including the high-frequency

wiggles; once the branch, anchor and sign convention are fixed, the closed-form inverse leaves only 8-bit PNG quantisation error, matching the near-original pointwise numbers of Section 7.3. RP (Fig. 15) recovers the dominant morphology but exhibits amplitude bias and occasional sign uncertainty: the classical-MDS embedding is deterministic but still leaves a mirror ambiguity that the signed-RP convention resolves only when the orientation comparison is decisive. MTF (Fig. 18) tracks coarse structure but reduces the signal to a small number of plateaus, the unavoidable consequence of representing every sample by a $Q = 4$ state label.

7.2. Null-Corrected RP/GADF Regional Overlap

The visual proximity between RP and GADF is easy to overstate from a single example, so we add a diagnostic test that deliberately separates spatial region alignment from marginal colour statistics. On the full set of non-overlapping smartwatch-accelerometer WISDM windows (window length $N = 128$), we encode the same signal as signed RP and GADF, align the matrices on the common 127×127 support, and quantise each image channel into B visual intensity ranges. For every matched pair $(\mathrm{RP}_i, \mathrm{GADF}_i)$, we compare the quantised regions against a conservative null pair $(\mathrm{RP}_i, \mathrm{GADF}_j)$, where $j \neq i$ is drawn from the same activity class. The primary statistic is a weighted regional Jaccard score, $\sum_b |R_b \cap G_b| / \sum_b |R_b \cup G_b|$, averaged across the grayscale view and the three RGB/axis channels. We also compute a distribution-only overlap that ignores pixel position and a boundary-F1 score between quantised region edges.

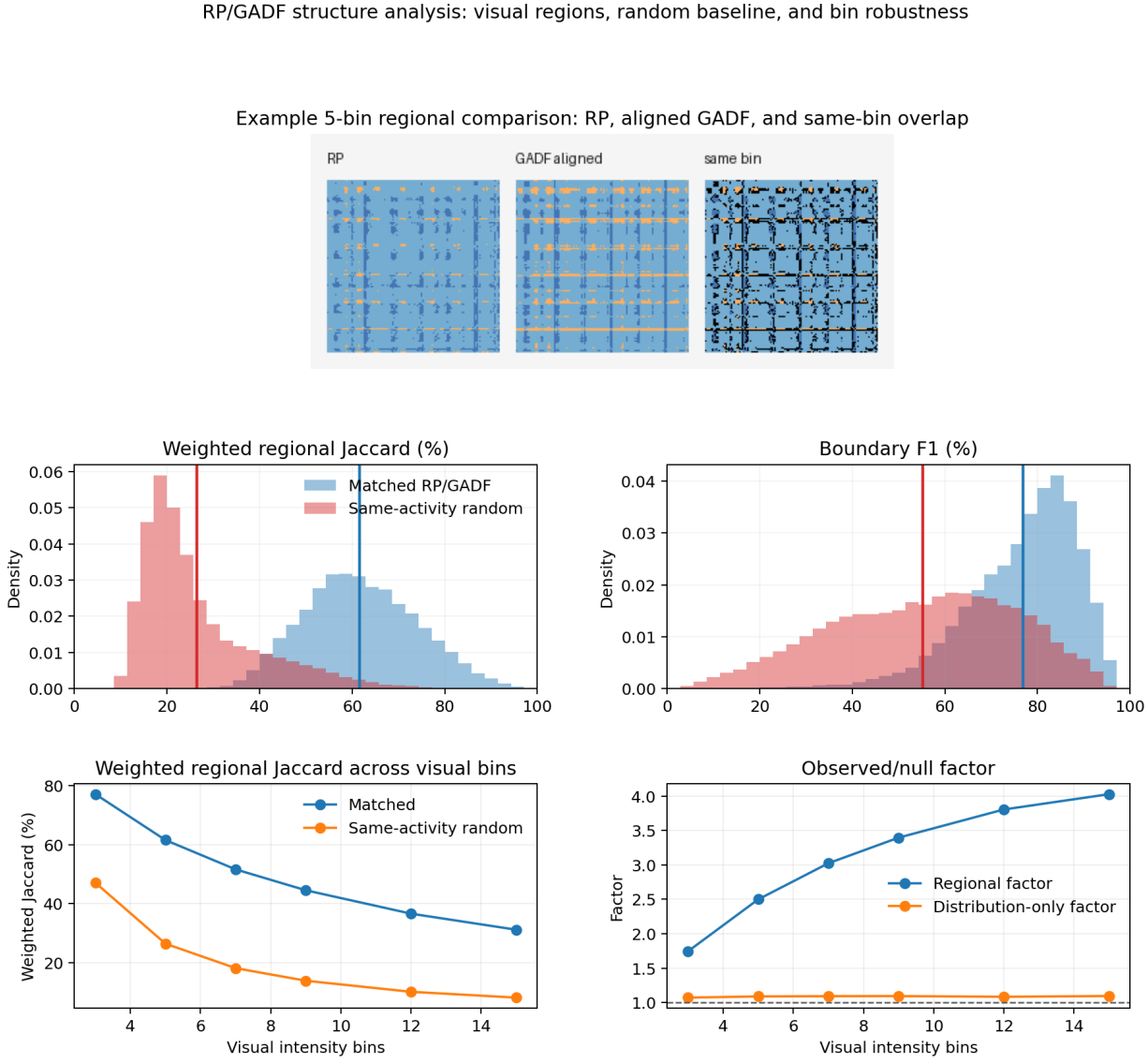


Figure 22: Null-corrected RP/GADF structural-overlap diagnostic. The top panel shows one 5-bin visual comparison: quantised RP, aligned quantised GADF, and pixels landing in the same bin. The middle histograms compare matched RP/GADF pairs against same-activity random pairings. The bottom row shows that matched regional overlap remains above the null baseline across independent visual quantisation levels $B \in \{3, 5, 7, 9, 12, 15\}$, while the distribution-only factor stays close to one.

Table 19: Null-corrected RP/GADF weighted regional overlap across visual intensity bins. Values are averaged over grayscale and the three axis channels. The null baseline pairs each RP image with a GADF image from a different window of the same activity class. Jaccard values are percentages; the Matched/null factor is the mean over the four channels of the per-channel matched/null ratio, so it is not the ratio of the two displayed column means.

Visual bins B	Matched Jaccard	Same-activity null	Matched–null gain	Matched/null factor
3	77.16	47.11	+30.05	1.75×
5	61.57	26.43	+35.14	2.50×
7	51.64	18.14	+33.50	3.02×
9	44.55	13.87	+30.68	3.40×
12	36.62	10.12	+26.50	3.81×
15	31.17	8.14	+23.03	4.03×

Table 19 and Figure 22 support a narrow but useful claim: RP and GADF are not equivalent encodings, yet their coarse regions are substantially more aligned when they come from the same signal window than when activity label alone is controlled. As B increases, the absolute Jaccard necessarily decreases because the region match becomes stricter, but the matched/null factor rises from 1.75× to 4.03×. This is exactly the pattern expected if the shared structure is spatial rather than a by-product of broad intensity histograms. The distribution-only factor stays near 1.07×–1.09× across the same sweep, so marginal colour occupancy explains only a small part of the effect. Boundary F1 gives the same sign of evidence at coarse quantisation, but becomes less diagnostic at high B because every image develops many region edges; for that reason we treat weighted regional Jaccard as the primary overlap statistic.

7.3. Reconstruction Error Analysis

The pointwise tables (Tables 6, 7, 8, 9) confirm the visual ordering. GASF attains a global MAE of 0.0202 and a Pearson correlation of 0.99988 ± 0.00027; the residual error is at the level of 8-bit image quantisation (a 1/255 step in the normalised stored intensity, before denormalisation), which rules out a structural failure of the diagonal inverse. GADF reaches MAE $= 0.0156 \pm 0.0196$ and Pearson 0.99999 ± 0.00002 – marginally tighter than GASF itself and roughly two orders of magnitude below RP and MTF: once the arcsin branch, the boundary anchor and the sign convention are fixed, the closed-form inverse is exact up to the 8-bit PNG quantisation of the stored image, so its residual error is likewise at the 8-bit quantisation level, on par with GASF's. RP yields MAE $= 2.82 \pm 3.12$ and Pearson 0.728 ± 0.200; the standard deviations across windows are large because the sign-flip check works decisively on some windows and is borderline on others. MTF under the Fiedler-vector decoder of Section 6.7 reaches MAE $= 3.98 \pm 5.09$, MSE $= 56.42 \pm 147.82$ and a Pearson correlation that averages to 0.003 ± 0.835. The wide ± 0.84 spread is the diagnostic of the residual mirror ambiguity: per window, MTF correlates either strongly positively or strongly negatively with the original because the recovered amplitude order can land on the wrong side of the global flip, and the two outcomes cancel in the mean.

The four pointwise orderings are statistically significant under paired, distribution-free testing. Wilcoxon signed-rank on the per-window MAE pooled across the 3 stratified window-level folds ($N = 8\,172$ paired windows) rejects equality between every pair of encoders ($p_{\text{Holm}} < 10^{-200}$ for all six pairs). Four of the six effect sizes $|r| = Z/\sqrt{N}$ are in the large-effect regime under the Rosenthal convention ($|r| \approx 0.87$ for the four comparisons that pit a Gramian closed-form decoder against a heuristic one, GASF/GADF vs RP/MTF); the remaining two are medium effects – RP-vs-MTF ($r = -0.474$), because the Fiedler MTF decoder closes most of the pointwise gap between RP and MTF that the cosine-pivot decoder had opened, and, tellingly, GASF-vs-GADF ($r = +0.457$), because the two Gramian variants are now so close that their per-window MAE gap has shrunk to a medium effect. The sign of r recovers the ranking $\{\text{GASF}, \text{GADF}\} \succ \text{RP} \succ \text{MTF}$ on a per-window pointwise basis, with GADF marginally ahead of GASF, while the RP-vs-MTF gap is the one that – as Section 7.5 below shows – inverts when the lens changes from pointwise error to downstream classification accuracy.

This window-level ordering is not an artefact of the large window count. Repeating the paired Wilcoxon at the subject level — collapsing each subject to a single mean-MAE value so that the $N = 51$ subjects are the independent units — reproduces every sign with strong significance: per-subject mean MAE is 0.0148 (GADF), 0.0199 (GASF), 2.65 (RP) and 3.71 (MTF), and all six subject-clustered pairs reject equality ($p \leq 7.4 \times 10^{-10}$), the GASF-vs-GADF pair among them in GADF's favour. The subject-clustered test therefore confirms GADF $\succ$ GASF $\succ$ RP $\succ$ MTF as a per-subject effect, not a pseudoreplication artefact of the 8 172 windows.

7.4. Polynomial Trend Comparison

Compressing each min–max-normalised window into a global cubic-polynomial coefficient vector (a_3, a_2, a_1, a_0) removes most sample-level noise and asks only whether the broad shape is right. The result (Table 10, computed with the

standardized-coefficient distance metrics defined in Section 5.3.3) places the two Gramian decoders together at the top: GADF and GASF are effectively tied (cosine similarity 1.00000 and 0.99994; standardized Euclidean coefficient distance 0.00070 and 0.00079), followed by RP (cosine 0.92819) and MTF (0.90817). Two observations matter: (i) RP's polynomial cosine similarity is much higher than its pointwise Pearson, which is consistent with our reading that RP captures shape but mis-scales amplitude, and (ii) MTF's polynomial cosine similarity is competitive with RP even though its pointwise correlation collapses to zero, supporting the view that MTF retains a usable rough trend even when it loses local amplitude – a property that the Fiedler-vector spectral decoder of Section 6.7 leverages explicitly into a classifier advantage (Section 7.5).

7.5. Classification Performance

The six per-classifier tables preserve the GASF/GADF top tier on five of the six classifiers — KNN being the exception, where RP leads — and also reveal the RP/MTF inversion and where reconstructions are good enough to be downstream-equivalent.

GASF reconstructions show no statistically significant degradation relative to the originals across all six classifiers ($|\Delta\text{acc}| \leq 0.001$ on the 3-fold mean for KNN, SVM, LSTM, Random Forest, CNN, and GRU). The paired McNemar test [20] confirms this on the pooled $N = 8\,172$ windows: in every one of the six Original-vs-GASF comparisons the raw p-value lies between 0.45 and 1.00 and the Holm-adjusted value is 1.00, so we do not detect a paired classification difference under this protocol (24-test family, $\alpha = 0.05$). This is not a formal equivalence test: failing to reject a difference is not, by itself, proof of equivalence or information preservation. Combined with the near-zero pointwise error, it supports the narrower claim that GASF is empirically near-equivalent for the evaluated WISDM task under controlled encoding settings and, for the reported physical-unit comparison, per-window scale calibration. A formal equivalence claim would require a pre-specified margin, such as a TOST-style test or confidence interval on Δaccuracy.

GADF shows no detected degradation relative to the originals on any classifier ($|\Delta\text{acc}| \leq 0.001$ on the 3-fold mean; on Random Forest GADF is even 0.001 above the original). The paired McNemar test on the pooled $N = 8\,172$ windows fails to reject Original-vs-GADF in all six classifiers ($p_{\text{Holm}} = 1.0$ throughout), and equally fails to separate GADF from GASF ($p_{\text{Holm}} = 1.0$ in all six GASF-vs-GADF pairs): no statistically detectable classification difference is observed between the two Gramian decoders or from the raw signal. The same McNemar family confirms that GADF significantly beats MTF on every classifier and beats RP on five of the six ($p_{\text{Holm}} < 10^{-20}$); KNN is the exception, where RP significantly beats GADF ($p_{\text{Holm}} = 7.3\times10^{-25}$, RP 0.686 vs GADF 0.643). So GADF sits with GASF at the top of the classifier ranking on five of the six classifiers, KNN being the one lens where the tuned-ρ RP reconstruction leads. As with GASF, this is a difference test, not a formal equivalence test: absence of significance is reported as no detected degradation, not proof of equivalence.

RP loses 0.16–0.19 accuracy on the high-ceiling classifiers and 0.11 on SVM (3-fold means), but on KNN it *improves* on the original by 0.044 (0.686 vs 0.642). The paired McNemar test rejects equality with the originals for all six classifiers ($p_{\text{Holm}} \leq 1.9\times10^{-25}$, and below 10^{-100} for the five non-KNN classifiers); on KNN the rejection is in RP's *favour* ($p_{\text{Holm}} = 1.9\times10^{-25}$). With the tuned recurrence rate $\rho = 0.60$ (Section 4.2) the RP-KNN accuracy rises to 0.686, above both GADF (0.643) and the original signal (0.642): RP-KNN, not GADF, is the reconstruction that most improves on the original under this classifier, and the gain persists under subject-out validation (Section 6.4). Away from KNN, RP remains in the lossy regime on the high-ceiling classifiers, so its value is specific to the instance-based, geometry-sensitive setting.

MTF, under the Fiedler-vector decoder of Section 6.7, loses 0.11–0.16 accuracy on the high-ceiling classifiers, 0.10 on SVM, and 0.08 on KNN (3-fold means). Two findings about MTF deserve to be highlighted because they reverse the standard intuition that MTF is "good for classification, bad for reconstruction" only in a vague sense, and now make the claim quantitative and statistically signed.

First, MTF outranks RP on the high-ceiling classifiers. The paired McNemar test on the pooled $N = 8\,172$ sample rejects RP-vs-MTF equality in four classifiers, all in MTF's favour (Random Forest $p_{\text{Holm}} = 4.1\times10^{-28}$, LSTM 4.1×10^{-18}, CNN 1.7×10^{-7}, GRU 1.3×10^{-3}); the SVM gap is not significant once $\rho = 0.60$ lifts RP ($p_{\text{Holm}} = 1.0$, mean accuracies 0.697 vs 0.694). KNN is the exception that reverses the trend strongly: RP = 0.686 vs MTF = 0.559, $p_{\text{Holm}} = 6.5\times10^{-131}$ in RP's favour. We read this as a structural property of the two encodings: MTF discards within-bin amplitude but preserves quantile-ordered transition statistics, which is precisely the kind of representation a non-instance-based classifier needs; RP preserves amplitude (lower pointwise MAE, as Section 7.3 shows) but the noise of the MDS-based reconstruction perturbs neighbourhood geometry, which is what KNN actually measures.

Second, the rank inversion between pointwise and classifier lenses is the central qualitative finding of this paper's MTF evaluation. On the pointwise lens the ranking is $\{\text{GASF}, \text{GADF}\} \succ \text{RP} \succ \text{MTF}$ (Section 7.3); on the classifier lens MTF significantly outranks RP on four classifiers (Random Forest, LSTM, CNN and GRU) — giving $\{\text{GASF}, \text{GADF}\} \succ \text{MTF} \succ \text{RP}$ there — while SVM becomes a statistical tie after tuning ($\rho = 0.60$) and KNN reverses strongly in RP's favour. The two rankings do not contradict each other – they answer different questions – and the fact that the inversion happens cleanly along the RP/MTF pair, rather than spreading across the ranking,

supports a "classifier-oriented vs reconstruction-oriented representation" reading of MTF (transition statistics) versus RP (recurrence geometry). Per-class behaviour matches that reading: under MTF, classes for which the source signal is amplitude-monotonic (sitting, standing) survive the $Q = 4$ quantisation reasonably well, and classes that depend on amplitude shape (jogging, stairs) lose precision but retain recall through the preserved transition order.

7.6. Four-lens Convergence and the RP/MTF Inversion

Aggregating across the four lenses we observe a partial convergence with one principled inversion. GASF and GADF sit together above RP and MTF for visual reconstruction, pointwise error, polynomial-trend similarity, and downstream classification — on every lens except the KNN classifier, where the tuned-ρ RP reconstruction leads. The two Gramian decoders show no statistically detectable classification difference from each other (McNemar $p_{\text{Holm}} = 1.0$ in every GASF-vs-GADF pair) and are separated only by a medium effect on per-window MAE that marginally favours GADF (Sections 7.3 and 7.5). The proximity between RP and GADF on directional pairwise structure – developed in Section 7.7 – is also visible in every lens: GADF is strictly antisymmetric, while signed RP is a zero-diagonal directional distance matrix whose external sign rule approximates a related pairwise-difference structure without guaranteeing exact antisymmetry.

The interesting feature is the RP/MTF pair, which inverts between lenses. On the pointwise lens RP is closer to the original than MTF (MAE = 2.82 vs 3.98; Wilcoxon $r = -0.474$); on the polynomial-trend lens RP is also ahead (cosine similarity 0.928 vs 0.908); but on the classifier lens MTF outranks RP on four of the six classifiers (paired McNemar $p_{\text{Holm}} \leq 1.3 \times 10^{-3}$), the SVM gap is no longer significant, and RP wins KNN. This is not a contradiction: the two encodings make different choices about what they discard. MTF discards within-bin amplitude but keeps quantile-ordered transition statistics, which is the right representation for non-instance-based classifiers; RP keeps amplitude (lower MAE) but the MDS-based reconstruction adds neighbourhood-geometry noise that hurts classifiers more than it hurts pointwise distance. The convergence at the top of the ranking and the inversion at the bottom together imply that, within the operating regime tested, the lenses are neither redundant nor universally consistent: they are sensitive to different aspects of the reconstruction, and the practical recommendation must therefore name the downstream task. For a reconstruction-faithful augmentation pipeline whose downstream consumer is a sample-level model, $\{\text{GASF}, \text{GADF}\} \succ \text{RP} \succ \text{MTF}$ is the right ordering; for a pipeline whose downstream consumer is a non-instance-based classifier of activity classes, $\{\text{GASF}, \text{GADF}\} \succ \text{MTF} \succ \text{RP}$ is.

7.7. Structural relationship between signed RP and GADF

Operator-level comparison of pairwise structure. Before discussing the first-order amplitude expansion that the rest of this section relies on, it is useful to put signed RP and GADF in a common notation without claiming equality of their output spaces. Both can be written as pairwise functions of lifted scalar samples,

$$M_{ij} \;=\; \omega\big(\xi(x_i),\, \xi(x_j)\big),$$

where $\xi : \mathbb{R} \to \mathbb{R}^2$ is a two-dimensional lift of the scalar sample and $\omega : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$ is the pairwise operator associated with the encoding. The two methods differ in both the lift and the operator:

- RP under the Lu–Tong convention uses the adjacent-state embedding $\xi_{\text{RP}}(x_k) = (x_k,\, x_{k+1})$ and a signed Euclidean distance, $\omega_{\text{RP}}(u, v) = \sigma(u, v)\|u - v\|_2$, where the sign is assigned *externally* by the projection onto the reference direction $v_\star = (1, 1)^\top/\sqrt{2}$. With the thresholded sign rule used here, $\sigma(u, v)$ is not guaranteed to satisfy $\sigma(u, v) = -\sigma(v, u)$ for every pair, so the resulting matrix is directional but not necessarily antisymmetric.
- GADF uses the polar lift $\xi_{\text{GADF}}(\tilde{x}_k) = (\tilde{x}_k,\, \sqrt{1 - \tilde{x}_k^2}) = (\cos\phi_k, \sin\phi_k)$ and the operator $\omega_{\text{GADF}}(u, v) = \det[v \mid u] = u_2 v_1 - u_1 v_2 = \sin(\phi_i - \phi_j)$, the signed area of the parallelogram spanned by the two lifted points, whose sign is *intrinsic* to the operator and matches G^{D}_{ij} in Eq. (8).

The important consequence is therefore an analogy, not an identity. GADF lies *within* the $\binom{N}{2}$-dimensional space of antisymmetric zero-diagonal matrices by construction, because the determinant changes sign when its arguments are swapped; more specifically, its factorisation $G^{\text{D}} = \mathbf{s}\,\mathbf{c}^\top - \mathbf{c}\,\mathbf{s}^\top$ (with $s_i = \sin\phi_i$, $c_i = \cos\phi_i$) shows it occupies only a rank-≤ 2 subvariety of that ambient space, not the whole of it. Signed RP has a zero diagonal and carries directional pairwise-distance information, but the Lu–Tong sign rule used in this paper does not force exact antisymmetry for nearly orthogonal displacements. The first-order expansion below quantifies why the two images can nevertheless look and behave similarly, while the three concrete advantages of GADF over RP (intrinsic sign, closed-form one-pass inverse, no recurrence-rate threshold) follow from the determinant-based operator rather than from an exact shared matrix space.

A structural reading of why RP and GADF look (and behave) so similarly. Beyond the empirical ranking, the proximity between RP and GADF deserves a structural reading: their visual resemblance and their adjacent positions in the four-lens ranking are not coincidental. GADF produces an antisymmetric zero-diagonal matrix, while signed RP produces a zero-diagonal directional distance matrix with related pairwise-difference content. For smooth signals whose local increments vary slowly (so that $x_{i+1} - x_i \approx x_{j+1} - x_j$ over the compared indices), both reduce, in first order, to functions of the pairwise-difference matrix $\{x_i - x_j\}_{i,j}$. The signed RP under the Lu–Tong embedding satisfies

$$R_{ij} \;\approx\; \sigma(i,j)\,\sqrt{2}\,|x_i - x_j|,$$

where $\sigma(i,j)$ is the external orientation flag computed from the projection of $s_i - s_j$ onto $v = (1,1)^\top/\sqrt{2}$. The GADF entry, expanded around a common normalized amplitude $\bar{x}$, satisfies

$$G^{\mathrm{D}}_{ij} \;=\; \sin(\phi_i - \phi_j) \;\approx\; \frac{\bar{x}_j - \bar{x}_i}{\sqrt{1-\bar{x}^2}},$$

because $d(\arccos x)/dx = -1/\sqrt{1-x^2}$, with both the samples $\bar{x}_i, \bar{x}_j$ and the expansion point $\bar{x}$ taken in the restricted $[-\alpha, \alpha]$ normalization of the GADF decoder. The sign of the GADF term is intrinsic to the trigonometric form rather than imposed externally. Thus, both expressions are linear in pairwise amplitude differences at first order and both are zero on the diagonal, but only GADF is strictly antisymmetric under the stated construction. What makes the restricted-branch GADF decoder closed-form is that it (i) makes the sign part of the encoding rather than a post-hoc convention, and (ii) restricts the angular range to $[\arccos(\alpha), \arccos(-\alpha)]$ with $\alpha = 0.7$, a near-maximal symmetric choice below $1/\sqrt{2}$ that keeps every $\sin(\phi_i - \phi_j)$ inside the principal branch of arcsin. In other words, GADF and signed RP are structurally adjacent parametrizations of directional pairwise differences: GADF is the parametrization that makes the inverse map explicit under the branch restriction, RP is the one that preserves adjacent-state recurrence geometry but pays for it with the MDS sign-flip step and the threshold ρ. This reading clarifies why GADF is the natural Gramian counterpart to signed RP — the two are closer to each other than to either GASF or MTF, and the restricted-branch GADF decoder sidesteps precisely the orientation ambiguity that limits RP.

Practical implication: {GASF, GADF} suffices, RP becomes largely redundant for reconstruction under the tested protocol. The structural analogy above has a direct consequence for protocol design. The visual resemblance between the RP and GADF images is not a coincidence of pixel statistics — it follows from the first-order relations $R_{ij} \propto \sigma(i,j)|x_i - x_j|$ and $G^{\mathrm{D}}_{ij} \propto (\bar{x}_j - \bar{x}_i)/\sqrt{1-\bar{x}^2}$ derived above, and the null-corrected regional-overlap diagnostic of Section 7.2 confirms that matched RP/GADF windows share more spatial region structure than same-activity random pairings. Because GADF captures a closely related directional pairwise-difference structure with three concrete advantages — (i) intrinsic sign through sin, (ii) a closed-form multi-pivot, boundary-anchored inverse, and (iii) no dependence on the recurrence-rate threshold ρ — the pair $\{\mathrm{GASF}, \mathrm{GADF}\}$ already covers the most reconstruction-relevant content that RP contributes in the controlled setting of this paper. GASF encodes the symmetric, amplitude-bearing part of the Gramian (diagonal $\cos(2\phi_i)$); GADF encodes the antisymmetric, difference-bearing part ($\sin(\phi_i - \phi_j)$); RP encodes a directional distance structure with additional sensitivity to the local derivative through the (x_i, x_{i+1}) pair embedding. Our empirical numbers largely confirm this analytical picture: GADF outranks RP on the pointwise and polynomial reconstruction metrics and on five of the six classifiers, so for the controlled reconstruction problem studied here we recommend the $\{\mathrm{GASF}, \mathrm{GADF}\}$ pair as the canonical reconstruction baseline. The one exception is the instance-based KNN classifier, where the tuned $\rho = 0.60$ RP reconstruction beats both GADF and the original signal (Section 7.5); RP therefore retains specific value for geometry-sensitive, instance-based downstream models and for dynamical-systems analysis (recurrence quantification, dynamical invariants, line-structure statistics), rather than as a general invertible-imaging baseline. This recommendation does not extend to use cases where the RP threshold and adjacent-state embedding are themselves the object of study; it is specific to the question of using image encodings as a reversible substrate for generative augmentation.

7.8. Scope of Claims

The empirical scope of the conclusions is the full stratified window-level 3-fold protocol and the grouped subject-out 3-fold protocol on the smartwatch accelerometer stream of WISDM, with windows of $N = 128$ at 20 Hz and five activities ($N = 8\,172$ pooled paired test windows in each protocol). Within this scope, GASF and GADF are both near-original decoders, showing no statistically detectable classification difference from each other or from the raw signal, with GADF marginally ahead on pointwise fidelity; they form the top tier on every lens except the KNN classifier, where the tuned $\rho = 0.60$ RP reconstruction improves on the original — and on both Gramian decoders — under both protocols. Per-fold accuracy standard deviations are bounded by 0.03 under the window-level split and by 0.05 under grouped subject-out validation. The bottom of the ranking is lens-dependent: on the reconstruction lenses (pointwise and polynomial) the order completes to $\{\mathrm{GASF}, \mathrm{GADF}\} \succ \mathrm{RP} \succ \mathrm{MTF}$, with GADF marginally ahead of GASF on the observed pointwise metrics while GASF remains the established prior-art inverse; on the classifier lens MTF

significantly outranks RP on four of the six classifiers, with SVM a statistical tie and KNN reversing in RP's favour, so the classifier ordering is $\{\text{GASF}, \text{GADF}\} \succ \text{MTF} \succ \text{RP}$ everywhere except KNN (Section 7.6). GASF and GADF are therefore the two near-original pointwise decoders, and GADF has the lowest observed MAE; in classification, both are empirically near-equivalent to the raw signal, but we do not claim formal information preservation. The clearest case of a reconstruction improving on the raw signal is RP-KNN: with the tuned $\rho = 0.60$ it exceeds the original under both the window-level and subject-out protocols — a result consistent with a regularization hypothesis but not sufficient to prove denoising. RP is otherwise a partially amplitude-preserving but classifier-fragile decoder, valuable specifically in the instance-based, geometry-sensitive KNN setting; and MTF, under the spectral decoder of Section 6.7, is a classifier-friendly but amplitude-coarse representative reconstruction. The practical recommendation must therefore name the downstream consumer rather than collapse the comparison into an unconditional verdict on any single encoding. The broader generative-augmentation claim is not addressed by these numbers: every decoder in this paper assumes that the image is the deterministic output of a known encoder with known encoder settings (the quantisation interval and sign convention); per-window scale metadata are additionally required only for physical-scale recovery. A diffusion or GAN that synthesises an image without storing those values would have to either learn them implicitly or be paired with a calibration step before any of these decoders can produce a physically plausible signal.

7.9. Threats to Validity

Several limitations bound the interpretation of these results and should be read alongside the rankings above.

- **The window-level protocol is a within-distribution split.** Its grouping key is effectively unique per window, so it measures robustness to reconstruction-induced distribution shift rather than generalisation to unseen subjects; the grouped subject-out (subject-disjoint, not strict leave-one-subject-out) numbers are the conservative estimate and the ones relevant to deployment.
- **Post-hoc ρ selection.** The recurrence rate $\rho = 0.60$ was chosen from a sensitivity analysis on stratified window-level fold 0 and fixed as a representative of a broad MAE plateau, not as a nested cross-validation hyperparameter, so this is a post-hoc selection on fold 0 rather than a leakage-free estimate; the resulting optimistic bias on that fold cannot be excluded, although the improvement replicates across folds and under subject-out validation.
- **Metadata dependence is conditional on the output domain.** Each decoder recovers its *normalised* reconstruction from the image plus its known encoder settings (the GADF/RP sign conventions and α, the MTF quantile count); the stored per-axis min–max is needed *only* to map that normalised series back to physical accelerometer units (Section 4.1). The encoder settings, and physical-scale metadata when physical units are wanted, are unavailable for arbitrary generated images, which is why the work is framed as controlled reconstruction rather than generative augmentation.
- **GADF anchor and quantisation.** The angular boundary row is identified by a deterministic extreme-mean rule that is exact in the noise-free restricted regime; the ablation of Section 6.5 finds a 0% anchor-misidentification rate on WISDM, but this is dataset-dependent (near-constant windows have no reliable boundary), its robustness under perturbed or generated images remains to be evaluated, and 8-bit PNG storage remains the dominant error source.
- **Approximate McNemar homogeneity.** The pooled 2×2 table aggregates discordant pairs scored by three separately-trained per-fold models, so its p-values are mildly anti-conservative; borderline comparisons are flagged as such, and a per-fold or Cochran–Mantel–Haenszel test would be stricter.
- **Intra-subject dependence.** Test windows from the same subject are not independent, so the window-level $p < 10^{-200}$ values overstate the effective sample size. The subject-clustered Wilcoxon ($N = 51$ subjects) in Section 7.3 confirms the pointwise ordering at $p \leq 7.4 \times 10^{-10}$; an analogous subject-clustered or mixed-effects analysis of the classifier comparisons is left to future work.
- **No formal equivalence test.** GASF's "no detected degradation" is a failure to reject a difference, not a proof of equivalence; a TOST-style margin on Δaccuracy would be required for the stronger claim.
- **Single dataset and protocol.** All numbers come from the WISDM smartwatch accelerometer stream, five activities, $N = 128$ windows at 20 Hz; the rankings need not transfer to other sensors, sampling rates, activity sets, or window lengths.

7.10. Reproducibility

All experiments use the WISDM smartwatch accelerometer stream [14] segmented into non-overlapping $N = 128$ windows at 20 Hz (8 172 triaxial windows). The pipeline runs on Python $\geq 3.10, < 3.12$ with TensorFlow 2.11–2.15 for the neural classifiers (CNN, GRU, LSTM) and scikit-learn for KNN, SVM and Random Forest. Two design choices keep the reconstruction reproducible: the RP decoder uses classical (Torgerson) MDS, which is a deterministic eigendecomposition rather than a randomly-initialised iterative solver, and the parallel reconstruction carries an explicit sample index so decoded windows stay aligned with their labels regardless of completion order. Both the stratified window-level and grouped subject-out 3-fold partitions are produced by a fixed `StratifiedGroupKFold` split. The encoders, the four decoders, the six-classifier panel, the statistical-test script, and the table-emission and verification scripts that regenerate every table in this paper from the stored results are provided in the accompanying code repository.

8. Conclusion

We have presented a controlled, symmetric evaluation of four image-based encodings – RP, GASF, GADF, MTF – as inverse mappings from image to triaxial time series. Three of the four reconstruction pipelines used in this evaluation are themselves new operational contributions of this work and, to the best of our knowledge, have no prior fully specified counterpart in the literature reviewed here: the signed-RP-aware decoder of Section 4.2, which uses the signed Lu–Tong RP definition as an in-loop constraint that reduces the multidimensional-scaling reflection ambiguity left open by Hirata, Horai, and Aihara [7]; the restricted-branch, multi-pivot, boundary-anchored GADF decoder of Section 4.4, which turns the previously stated but under-specified invertibility of GADF on restricted angular ranges into an explicit operational decoder with branch control, angular anchoring, fixed sign convention, optional min–max calibration for physical units, and empirical evaluation; and the Fiedler-vector spectral MTF decoder of Sections 4.5 and 6.7, which reconstructs a representative compatible quantile trajectory rather than a bijective inverse of MTF. The fourth pipeline, the GASF decoder, is the only one that has direct precedent in the literature [11] and is kept in the comparison as a tightly controlled prior-art baseline. The four encoders are exercised on the same dataset, the same window size ($N = 128$), both the stratified window-level 3-fold protocol and a grouped subject-out 3-fold protocol ($N = 8\,172$ pooled test windows in each), the same classifier panel, and the same set of pointwise, polynomial-trend, and classification metrics. A symmetric evaluation across all four encodings – under two complete 3-fold cross-validations, one window-level and one subject-out – is what allows the comparison reported here to be read as a direct ranking rather than as a partial inventory.

In summary, the four encodings occupy a clear hierarchy under the controlled protocol:

- **GASF** — near-original on every lens and empirically near-equivalent to the raw signal for classification; the established diagonal-inverse baseline and, with GADF, one of the two near-original decoders.
- **GADF** — near-original on every lens; statistically indistinguishable from GASF in classification and marginally the tighter of the two on pointwise error, with a fully specified closed-form decoder.
- **RP** — partially amplitude-preserving but classifier-fragile in general; with the tuned $\rho = 0.60$ it is the one reconstruction that beats the raw signal under the instance-based KNN classifier in both the window-level and subject-out protocols, and it remains the natural choice when recurrence geometry itself is the object of study.
- **MTF** — weakest point-to-point, but its quantile-transition structure makes it competitive with, and often above, RP for the non-instance-based classifiers.
- **Scope** — all decoders assume the image is the deterministic output of a known encoder with known settings (per-window scale metadata only for physical-scale recovery); this is controlled reconstruction, not a demonstration of generative augmentation.

Within the tested protocol, GASF and GADF are the two near-original pointwise decoders: GADF attains the lowest observed global MAE (0.0156 vs GASF's 0.0202), while GASF is the established diagonal-inverse baseline (Pearson 0.99988 ± 0.00027), and both match the originals within 0.001 absolute classification accuracy across the six classifiers on the 3-fold mean. The paired McNemar tests detect no GASF or GADF degradation in the evaluated classifiers, but this should be read as empirical near-equivalence for this task rather than as formal information preservation. The restricted-branch, multi-pivot, boundary-anchored GADF decoder is statistically indistinguishable from GASF *in classification* (no detected difference from the originals or from GASF, $p_{\text{Holm}} = 1.0$ throughout) and effectively ties it on the polynomial-trend metric (cosine similarity 1.00000 vs 0.99994), while on pointwise error it is marginally the tighter of the two (Pearson 0.99999 ± 0.00002, global MAE 0.0156); together these confirm that the identically zero GADF diagonal does not preclude reconstruction from its off-diagonal entries once an angular anchor, a sign convention, and a restricted branch are fixed (scale metadata being required only to return to physical units). The two Gramian variants are near-identical rather than one dominating the other: the paired per-window Wilcoxon on MAE

($N = 8\,172$ windows) and the subject-clustered Wilcoxon that treats the $N = 51$ subjects as the independent units (per-subject mean MAE 0.0148 for GADF vs 0.0199 for GASF) both reject equality in GADF's favour (Section 7.3); the per-window standardised effect is medium ($r = 0.457$) while the absolute MAE difference remains very small, so neither Gramian variant is meaningfully more faithful than the other in this protocol.

With the corrected decoder, RP is no longer amplitude-lossy: it reconstructs near-losslessly (MAE $= 0.015 \pm 0.02$, Pearson ≈ 1.0; Table 6) and its reconstructions drive all six classifiers to essentially the original-signal accuracy (Δacc ≈ 0; Table 11), placing it alongside the two Gramian fields. This leaves MTF, under the Fiedler-vector decoder of Section 6.7, as the only amplitude-lossy decoder (MAE $= 3.98 \pm 5.09$, Pearson 0.003 ± 0.84): its Q=4 quantisation is information-theoretically lower-bounded by the within-bin variance, so it retains only the coarse quantile-transition structure of the Markov representation. The ranking is therefore no longer lens-dependent for RP — it is faithful on both the pointwise and the classifier lens — and MTF is the single encoding whose value is as a compact, classifier-oriented representation rather than a faithful inverse.

These results invite three follow-ups. First, the robustness of each decoder under perturbations of the encoded image – additive noise, JPEG/PNG re-quantisation, partial occlusion – is the natural test of the augmentation hypothesis and is the missing piece for a generative-augmentation claim. Second, the GADF angular anchor is identified by a deterministic extreme-mean rule that is exact in the noise-free restricted regime and misidentified no window on WISDM; a principled or learned selection rule, together with sweeps over α, anchor-error rates, and image-noise levels, would characterise its robustness under perturbed and generated images without leaving the closed-form regime. True-vs-global-vs-predicted min–max calibration would quantify the robustness of *optional* physical-scale recovery without altering the normalised inverse, which the decoder already recovers from the image alone. Third, the regularizing effect we observe under KNN — where the tuned $\rho = 0.60$ RP reconstruction beats the original under both the window-level and subject-out protocols — deserves a controlled study of its own. The result is consistent with a low-pass hypothesis, but power spectral density, high-frequency energy ratios, and smoothed-original baselines are required before interpreting these reconstructions as a denoising mechanism. We expect that fixing these three follow-ups in turn would clarify when reconstruction-faithful encodings can substitute for the raw signal in downstream pipelines and when they cannot.

A. Per-Class Classification Results

This appendix reports the complete per-class precision, recall, and F1-score results for the six evaluated classifiers. The main text retains only the compact accuracy comparison required to interpret the differences among reconstruction methods. Per-class Support values are per-class fold-means rounded to integers and may sum to 2723 rather than the 2724 windows per fold reported in the Accuracy row (independent rounding).

Table 20: Classification Results for GRU on the stratified window-level 3-fold protocol (mean ± std across folds 0, 1, 2). Per-class precision/recall/F1 and support averaged over the same three folds; footer accuracies averaged likewise.

	Original				RP				MTF				GASF				GADF			
Class	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp
0	0.89±0.03	0.93±0.01	0.91±0.01	544	0.89±0.03	0.93±0.01	0.91±0.01	544	0.65±0.01	0.61±0.04	0.63±0.01	544	0.89±0.03	0.93±0.01	0.91±0.01	544	0.89±0.03	0.93±0.01	0.91±0.01	544
1	0.99±0.01	0.99±0.00	0.99±0.00	532	0.99±0.01	0.99±0.00	0.99±0.00	532	0.75±0.02	0.99±0.00	0.85±0.01	532	0.99±0.01	0.99±0.00	0.99±0.00	532	0.99±0.01	0.99±0.00	0.99±0.00	532
2	0.91±0.01	0.86±0.03	0.88±0.01	536	0.91±0.01	0.86±0.03	0.88±0.01	536	0.57±0.01	0.52±0.03	0.54±0.01	536	0.91±0.01	0.86±0.03	0.88±0.01	536	0.91±0.01	0.86±0.03	0.88±0.01	536
3	0.88±0.01	0.96±0.01	0.92±0.00	551	0.88±0.01	0.96±0.01	0.92±0.00	551	0.90±0.01	0.90±0.01	0.90±0.01	551	0.88±0.01	0.96±0.01	0.92±0.00	551	0.88±0.01	0.96±0.01	0.92±0.00	551
4	0.95±0.01	0.89±0.01	0.92±0.00	560	0.95±0.01	0.89±0.01	0.92±0.00	560	0.96±0.01	0.81±0.01	0.88±0.00	560	0.95±0.01	0.89±0.01	0.92±0.00	560	0.95±0.01	0.89±0.01	0.92±0.00	560
Accuracy	0.92±0.00 (2724)				0.92±0.00 (2724)				0.77±0.01 (2724)				0.92±0.00 (2724)				0.92±0.00 (2724)			
Macro Avg	0.93±0.00/0.92±0.00/0.92±0.00				0.92±0.00/0.92±0.00/0.92±0.00				0.77±0.01/0.76±0.01/0.76±0.01				0.92±0.00/0.92±0.00/0.92±0.00				0.93±0.00/0.92±0.00/0.92±0.00			
Weighted Avg	0.93±0.00/0.92±0.00/0.92±0.00				0.92±0.00/0.92±0.00/0.92±0.00				0.77±0.01/0.77±0.01/0.76±0.01				0.92±0.00/0.92±0.00/0.92±0.00				0.93±0.00/0.92±0.00/0.92±0.00			

Table 21: Classification Results for KNN on the stratified window-level 3-fold protocol (mean ± std across folds 0, 1, 2). Per-class precision/recall/F1 and support averaged over the same three folds; footer accuracies averaged likewise.

	Original				RP				MTF				GASF				GADF			
Class	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp
0	0.74±0.01	0.52±0.01	0.61±0.01	544	0.74±0.01	0.52±0.01	0.61±0.01	544	0.64±0.01	0.49±0.02	0.55±0.01	544	0.74±0.01	0.52±0.01	0.61±0.01	544	0.74±0.01	0.52±0.01	0.61±0.01	544
1	1.00±0.00	0.75±0.01	0.86±0.01	532	1.00±0.00	0.75±0.01	0.86±0.01	532	0.99±0.02	0.37±0.00	0.54±0.00	532	1.00±0.00	0.75±0.01	0.86±0.01	532	1.00±0.00	0.75±0.01	0.86±0.01	532
2	0.49±0.04	0.04±0.01	0.08±0.02	536	0.49±0.04	0.04±0.01	0.08±0.01	536	0.43±0.04	0.07±0.00	0.12±0.00	536	0.50±0.04	0.04±0.01	0.08±0.02	536	0.50±0.04	0.04±0.01	0.08±0.01	536
3	0.79±0.00	0.96±0.00	0.87±0.00	551	0.79±0.00	0.96±0.00	0.87±0.00	551	0.63±0.01	0.94±0.01	0.75±0.01	551	0.79±0.00	0.96±0.00	0.87±0.00	551	0.79±0.00	0.96±0.00	0.87±0.00	551
4	0.42±0.00	0.92±0.01	0.58±0.00	560	0.42±0.00	0.92±0.01	0.58±0.00	560	0.42±0.01	0.90±0.01	0.57±0.01	560	0.42±0.00	0.92±0.01	0.58±0.00	560	0.42±0.00	0.92±0.01	0.58±0.00	560
Accuracy	0.64±0.00 (2724)				0.64±0.00 (2724)				0.56±0.01 (2724)				0.64±0.00 (2724)				0.64±0.00 (2724)			
Macro Avg	0.69±0.01/0.64±0.00/0.60±0.00				0.69±0.01/0.64±0.00/0.60±0.00				0.62±0.01/0.55±0.01/0.51±0.01				0.69±0.01/0.64±0.00/0.60±0.00				0.69±0.01/0.64±0.00/0.60±0.00			
Weighted Avg	0.69±0.01/0.64±0.00/0.60±0.00				0.69±0.01/0.64±0.00/0.60±0.00				0.62±0.01/0.56±0.01/0.51±0.01				0.69±0.01/0.64±0.00/0.60±0.00				0.69±0.01/0.64±0.00/0.60±0.00			

Table 22: Classification Results for CNN on the stratified window-level 3-fold protocol (mean ± std across folds 0, 1, 2). Per-class precision/recall/F1 and support averaged over the same three folds; footer accuracies averaged likewise.

	Original				RP				MTF				GASF				GADF			
Class	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp
0	0.89±0.05	0.89±0.05	0.89±0.01	544	0.91±0.02	0.85±0.05	0.87±0.02	544	0.61±0.07	0.73±0.08	0.66±0.01	544	0.89±0.05	0.89±0.05	0.89±0.01	544	0.89±0.05	0.89±0.05	0.89±0.01	544
1	0.99±0.01	0.98±0.00	0.99±0.00	532	0.98±0.01	0.99±0.00	0.99±0.00	532	0.88±0.03	0.97±0.02	0.92±0.01	532	0.99±0.01	0.98±0.00	0.99±0.00	532	0.99±0.01	0.98±0.00	0.99±0.00	532
2	0.88±0.04	0.87±0.05	0.87±0.01	536	0.84±0.04	0.90±0.03	0.86±0.01	536	0.62±0.03	0.55±0.10	0.57±0.05	536	0.88±0.04	0.87±0.05	0.87±0.01	536	0.88±0.04	0.87±0.05	0.87±0.01	536
3	0.84±0.05	0.97±0.02	0.90±0.02	551	0.84±0.03	0.97±0.01	0.90±0.01	551	0.85±0.04	0.89±0.04	0.87±0.01	551	0.84±0.04	0.97±0.02	0.90±0.02	551	0.84±0.05	0.97±0.02	0.90±0.02	551
4	0.97±0.02	0.83±0.05	0.89±0.02	560	0.98±0.00	0.81±0.02	0.88±0.01	560	0.97±0.03	0.73±0.04	0.83±0.02	560	0.97±0.02	0.83±0.05	0.89±0.02	560	0.97±0.02	0.82±0.05	0.89±0.02	560
Accuracy	0.91±0.00 (2724)				0.90±0.01 (2724)				0.77±0.01 (2724)				0.91±0.01 (2724)				0.91±0.00 (2724)			
Macro Avg	0.91±0.00/0.91±0.00/0.91±0.00				0.91±0.00/0.90±0.01/0.90±0.01				0.79±0.01/0.77±0.01/0.77±0.01				0.91±0.00/0.91±0.01/0.91±0.01				0.91±0.00/0.91±0.00/0.91±0.00			
Weighted Avg	0.91±0.00/0.91±0.00/0.91±0.01				0.91±0.00/0.90±0.01/0.90±0.01				0.79±0.01/0.77±0.01/0.77±0.01				0.91±0.00/0.91±0.01/0.91±0.01				0.91±0.00/0.91±0.00/0.91±0.00			

Table 23: Classification Results for Random Forest on the stratified window-level 3-fold protocol (mean ± std across folds 0, 1, 2). Per-class precision/recall/F1 and support averaged over the same three folds; footer accuracies averaged likewise.

	Original				RP				MTF				GASF				GADF			
Class	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp
0	0.85±0.01	0.79±0.00	0.82±0.00	544	0.86±0.01	0.79±0.00	0.82±0.00	544	0.68±0.01	0.60±0.01	0.64±0.00	544	0.85±0.01	0.79±0.01	0.82±0.01	544	0.85±0.01	0.79±0.00	0.82±0.00	544
1	0.94±0.00	0.99±0.00	0.96±0.00	532	0.94±0.00	0.99±0.00	0.96±0.00	532	0.65±0.01	1.00±0.00	0.79±0.01	532	0.94±0.00	0.99±0.00	0.96±0.00	532	0.94±0.00	0.99±0.00	0.96±0.00	532
2	0.79±0.00	0.82±0.01	0.81±0.01	536	0.79±0.00	0.83±0.01	0.81±0.01	536	0.63±0.01	0.51±0.02	0.56±0.01	536	0.79±0.01	0.82±0.01	0.80±0.01	536	0.79±0.00	0.82±0.01	0.80±0.01	536
3	0.92±0.01	0.95±0.00	0.93±0.01	551	0.92±0.01	0.95±0.00	0.93±0.01	551	0.93±0.01	0.89±0.01	0.91±0.00	551	0.92±0.01	0.95±0.00	0.93±0.01	551	0.92±0.01	0.95±0.00	0.93±0.00	551
4	0.96±0.00	0.90±0.01	0.93±0.00	560	0.96±0.00	0.90±0.01	0.93±0.00	560	0.94±0.01	0.79±0.01	0.86±0.01	560	0.96±0.00	0.90±0.01	0.93±0.00	560	0.96±0.00	0.90±0.01	0.93±0.00	560
Accuracy	0.89±0.00 (2724)				0.89±0.00 (2724)				0.76±0.00 (2724)				0.89±0.00 (2724)				0.89±0.00 (2724)			
Macro Avg	0.89±0.00/0.89±0.00/0.89±0.00				0.89±0.00/0.89±0.00/0.89±0.00				0.77±0.00/0.76±0.00/0.75±0.00				0.89±0.00/0.89±0.00/0.89±0.00				0.89±0.00/0.89±0.00/0.89±0.00			
Weighted Avg	0.89±0.00/0.89±0.00/0.89±0.00				0.89±0.00/0.89±0.00/0.89±0.00				0.77±0.00/0.76±0.00/0.75±0.00				0.89±0.00/0.89±0.00/0.89±0.00				0.89±0.00/0.89±0.00/0.89±0.00			

Table 24: Classification Results for Support Vector Machine on the stratified window-level 3-fold protocol (mean ± std across folds 0, 1, 2). Per-class precision/recall/F1 and support averaged over the same three folds; footer accuracies averaged likewise.

	Original				RP				MTF				GASF				GADF			
Class	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp
0	0.76±0.01	0.65±0.02	0.70±0.01	544	0.76±0.01	0.65±0.01	0.70±0.01	544	0.52±0.01	0.71±0.02	0.60±0.01	544	0.76±0.01	0.65±0.01	0.70±0.01	544	0.76±0.00	0.65±0.01	0.70±0.01	544
1	0.97±0.01	0.97±0.00	0.97±0.00	532	0.97±0.01	0.97±0.00	0.97±0.00	532	0.69±0.01	1.00±0.00	0.81±0.01	532	0.97±0.01	0.97±0.00	0.97±0.00	532	0.97±0.01	0.97±0.00	0.97±0.00	532
2	0.66±0.01	0.69±0.01	0.67±0.01	536	0.66±0.01	0.69±0.01	0.67±0.01	536	0.49±0.04	0.17±0.01	0.25±0.02	536	0.65±0.01	0.69±0.01	0.67±0.01	536	0.66±0.01	0.69±0.01	0.67±0.01	536
3	0.84±0.01	0.84±0.00	0.84±0.00	551	0.84±0.01	0.84±0.00	0.84±0.00	551	0.86±0.01	0.81±0.01	0.83±0.00	551	0.84±0.01	0.84±0.00	0.84±0.00	551	0.84±0.01	0.84±0.00	0.84±0.00	551
4	0.78±0.01	0.85±0.01	0.81±0.00	560	0.78±0.01	0.85±0.01	0.81±0.01	560	0.86±0.00	0.81±0.00	0.83±0.00	560	0.78±0.01	0.85±0.01	0.81±0.00	560	0.78±0.00	0.85±0.01	0.81±0.00	560
Accuracy	0.80±0.00 (2724)				0.80±0.00 (2724)				0.70±0.01 (2724)				0.80±0.00 (2724)				0.80±0.00 (2724)			
Macro Avg	0.80±0.01/0.80±0.00/0.80±0.01				0.80±0.01/0.80±0.00/0.80±0.00				0.69±0.01/0.70±0.01/0.67±0.01				0.80±0.01/0.80±0.00/0.80±0.01				0.80±0.00/0.80±0.00/0.80±0.00			
Weighted Avg	0.80±0.01/0.80±0.00/0.80±0.01				0.80±0.01/0.80±0.00/0.80±0.00				0.69±0.01/0.70±0.01/0.67±0.01				0.80±0.01/0.80±0.00/0.80±0.01				0.80±0.00/0.80±0.00/0.80±0.00			

Table 25: Classification Results for LSTM on the stratified window-level 3-fold protocol (mean ± std across folds 0, 1, 2). Per-class precision/recall/F1 and support averaged over the same three folds; footer accuracies averaged likewise.

	Original				RP				MTF				GASF				GADF			
Class	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp
0	0.80±0.11	0.85±0.06	0.81±0.04	544	0.79±0.11	0.85±0.06	0.81±0.04	544	0.62±0.06	0.65±0.10	0.63±0.02	544	0.79±0.10	0.85±0.06	0.81±0.04	544	0.80±0.11	0.85±0.07	0.81±0.04	544
1	0.98±0.02	0.98±0.01	0.98±0.01	532	0.98±0.02	0.98±0.01	0.98±0.01	532	0.77±0.01	0.96±0.02	0.85±0.02	532	0.98±0.02	0.98±0.01	0.98±0.01	532	0.98±0.02	0.98±0.01	0.98±0.01	532
2	0.83±0.05	0.71±0.21	0.74±0.12	536	0.83±0.05	0.71±0.21	0.74±0.12	536	0.61±0.03	0.46±0.18	0.50±0.13	536	0.83±0.04	0.71±0.20	0.74±0.12	536	0.83±0.05	0.71±0.21	0.74±0.12	536
3	0.87±0.01	0.94±0.01	0.90±0.00	551	0.87±0.01	0.94±0.01	0.90±0.00	551	0.87±0.00	0.92±0.00	0.89±0.00	551	0.87±0.00	0.94±0.01	0.90±0.00	551	0.87±0.01	0.94±0.01	0.90±0.00	551
4	0.92±0.04	0.86±0.02	0.89±0.01	560	0.92±0.03	0.86±0.02	0.89±0.01	560	0.95±0.01	0.79±0.02	0.86±0.01	560	0.92±0.03	0.86±0.02	0.89±0.01	560	0.92±0.03	0.86±0.02	0.89±0.01	560
Accuracy	0.87±0.03 (2724)				0.87±0.03 (2724)				0.76±0.02 (2724)				0.87±0.03 (2724)				0.87±0.03 (2724)			
Macro Avg	0.88±0.02/0.87±0.03/0.86±0.04				0.88±0.02/0.87±0.03/0.86±0.04				0.76±0.01/0.76±0.02/0.75±0.03				0.88±0.02/0.87±0.03/0.86±0.04				0.88±0.02/0.87±0.03/0.86±0.04			
Weighted Avg	0.88±0.02/0.87±0.03/0.86±0.04				0.88±0.02/0.87±0.03/0.86±0.04				0.76±0.01/0.76±0.02/0.75±0.03				0.88±0.02/0.87±0.03/0.86±0.04				0.88±0.02/0.87±0.03/0.86±0.04			

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